MIL-F-22606B(SHIPS)
12 June 1964
SUPERSEDING
MIL-F-22606A(SHIPS)
30 July 1962
(See 6.5)

MILITARY SPECIFICATION

FLASK COMPRESSED GAS AND END PLUGS FOR AIR, OXYGEN AND NITROGEN

1. SCOPE

1.1 Scope.- This specification covers steel flasks and end plugs for storing compressed gas (air, oxygen and nitrogen).

1.2 Classification.- Flasks and end plugs shall be of the following types, classes and services as specified (see 6.2).

Types
- Type GF - General service flask (for end plugs see table III).
- Type SF - Separator flask (also referred to as "moisture separator"), service A only (for end plugs see table III).
- Type CD - General service flask, curved design (for end plugs see table III).

Class
- Class 3000 - 3000 pounds per square inch (p.s.i.) design working pressure.
- Class 5000 - 5000 p.s.i. design working pressure.

Service
- Service A - Compressed air.
- Service B - Oxygen and oil-free nitrogen.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

FSC 8120

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SPECIFICATIONS

FEDERAL

QQ-P-416 - Plating, Cadmium (Electro Deposited).

MILITARY

MIL-P-15328 - Primer Pretreatment (Formula No. 117 for Metals).
MIL-P-15930 - Primer Coating, Shipboard, Vinyl-Zinc Chromate (Formula No. 120 for Hot Spray).
MIL-S-16216 - Steel Plate, Alloy, Structural, High Yield Strength (Hy-80 and Hy-100).
MIL-S-23009 - Steel Forgings, Alloy, High Yield Strength (Hy-80)
MIL-G-23652 - Gasket and Packing Material Petroleum and Phosphate Ester Fluid Resistant.

STANDARDS

FEDERAL

FED-STD-151 - Metals, Test Methods.

MILITARY

MIL-STD-271 - Non-Destructive Testing Requirements for Metals.
MIL-STD-278 - Welding and Allied Processes for Machinery for Ships of the United States Navy.
MIL-STD-418 - Mechanical Tests for Welded Joints.

PUBLICATIONS

BUREAU OF SHIPS

NAVSHIPS 250-692-2 - ~~Radiographic Standards~~ X-Rays for Production and Repair Welds.
NAVSHIPS 250-634-3 - Method for Computing Drop-Weight Tests to Determine NIL-Ductility Transition Temperature of Ferritic Steels.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS (NBS)
Handbook H28 - Screw Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D.C.)

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, New York.)

AMERICAN SOCIETY FOR TESTING MATERIALS (ASTM)
A336 - Alloy Steel Seamless Drum Forgings.

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia 3, Pennsylvania.)

AMERICAN IRON AND STEEL INSTITUTE (AISI)
Steel Products Manual.

(Application for copies should be addressed to the American Iron and Steel Institute, 350 Fifth Avenue, New York 1, New York.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Material.-

3.1.1 Flasks.- The material for flasks shall be steel in accordance with class F-22 of ASTM A336 or Hy-80 in accordance with MIL-S-23009 or MIL-S-16216, as applicable, having mechanical properties equal to those specified in table I.

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Table I - Mechanical properties.

Yield strength, at 0.2 percent offset	90,000 p.s.i. minimum
Tensile strength	110,000 p.s.i. minimum
Reduction in area	45.0 percent minimum
Elongation	15.0 percent in 2 inches
Charpy V-notch at minus 120°F. 1/	50 foot pounds
NDT (Nil ductility transition) temperature 2/	0°F. maximum

1/ Charpy V-notch test applies to Hy-80 steel only.

2/ NDT, as determined drop weight testing, applies to A336 steel only.

3.1.2 End plugs.- End plugs shall be forged steel either in accordance with the steel used in the fabrication of the flask body or steel in accordance with AISI type 4130 having tensile and yield strengths equal to the steel used in fabrication of the flask. Configuration of plugs shall be as shown in figures 1, 2, 3, and 4 unless otherwise approved by the bureau or agency concerned.

3.1.3 O-rings and backup rings.- Backup rings shall be made of teflon. O-rings shall be in accordance with type II of MIL-G-23652 top visual quality. Dimensions shall conform to figure 5.

3.2 Mechanical properties.- The mechanical properties of all portions of the flasks shall conform to table I.

3.3 Dimensional characteristics.- The dimensional characteristics of seamless and welded flasks shall be as shown in table II. Unless otherwise specified herein the difference between the major and minor diameters shall not exceed 2 percent of the outside diameter (o.d.) of the flask.

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Table II - Dimensional characteristics.

Class and type	Capacity (min.)	Outside diameter ± 3/32 inch	Length (+2 inches -0 inch)	Wall thickness (min.)	Weight (approx.)	Hydro-test pressure	Applicable figure
	Cu. ft.	Inches	Feet-Inches	Inches	Pounds		
Class 3000							
Type GF	4.0	18	3 - 11	.561	450	5000	6
Type GF	6.0	18	5 - 3	.561	615	5000	6
Type GF	8.0	18	6 - 7	.561	800	5000	6
Type GF	10.0	18	7 - 10	.561	915	5000	6
Type GF	15.5	18	11 - 6-1/2	.561	1350	5000	6
Type GF	21.0	18	15 - 2	.561	1770	5000	6
Type CD	21.0	18	15 - 2	.561	1770	5000	7
Class 5000							
Type SF	.5	6-5/8	3 - 10	1/.494	140	8330	1/8
Type GF	1.5	10-3/4	4 - 2	1/0.572	305	8330	6
Type GF	4.0	18	4 - 3	.933	850	8330	6
Type GF	6.0	18	5 - 8-1/2	.933	1135	8330	6
Type GF	8.0	18	7 - 2	.933	1415	8330	6
Type GF	10.0	18	8 - 8	.933	1705	8330	6
Type GF	10.0	20	7 - 4	1.028	1715	8330	6
Type GF	15.5	18	12 - 8-1/2	.933	2480	8330	6
Type GF	21.0	18	16 - 9	.933	3260	8330	6
Type CD	21.0	18	16 - 9	.933	3260	8330	7
Type GF	25.0	20	16 - 3	1.028	3820	8330	6
Type CD	25.0	20	16 - 3	1.028	3820	8330	7

1/ All 1/2 cubic foot capacity separators (type SF) and 1.5 cubic foot (type GF) flasks shall be class 5000 and shall be applicable to all pressures up to and including 5000 p.s.i. (see 6.4).

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3.4 Fabrication.- The body of types GF and SF flasks shall be in the shape of a straight cylinder as shown on figures 6 and 8. The type CD flask shall be a curved cylinder as shown on figure 7. Each end shall be hot formed to provide a neck for the flask end connections (see figure 9). Flasks may be fabricated of either seamless tubular material, pierced or drawn billets, or plate welded prior to final heat treatment. Welding for fabrication of flasks shall conform to the requirements of MIL-STD-278. The suitability of the electrode and procedure shall be established by making a double-butt weld specimen in material of the same analysis and wall thickness of the flask it represents and after welding the specimen shall be subjected to the equivalent heat treatment contemplated for the work. If final heat treatment requires quenching from one side only, the specimen shall have ends closed before heat treating. All main welded joints including welds for neck attachment shall be of the double welded butt type or its equivalent. Necks shall be flush with the interior and shall have a 1/2 inch minimum fillet weld on the outside. Where a backing strip is used, it shall be removed and the joint shall be examined by magnetic particle method in accordance with MIL-STD-271 prior to the required full radiographing. Welding construction of flasks is subject to the requirements of 6.3.

3.4.1 Heat treatment.- After all hot forming operations, including welding of long seams except seal welding, have been completed the flask shall be heated to above the upper critical transformation temperature and then quenched and tempered to produce a tempered martensitic structure. The minimum tempering temperature shall be 1100 degrees F. After heat treatment all welds shall be examined by the magnetic particle method in accordance with MIL-STD-271 and all main welded joints shall be radiographed. Defects indicated by radiograph shall not exceed group I of NAVSHIPS 250-692-2. Welding for repair of joints on flasks which are to be or have been heat treated may be made only after approval by the inspector. Defects shall be removed to sound metal as shown by acid etch, dye penetrant, or equivalent examination. After repair has been made the flask shall be again fully heat treated and all main joints magnetic particle examined and the affected repair area radiographed.

3.4.1.1 Stress relief.- When stress relieving of welds is required the stress relieving temperature shall not be higher than 100 degrees below the minimum specified tempering temperature.

3.4.2 End plugs.- Unless otherwise specified (see 6.2) all flasks shall be furnished with the end plugs as specified in table III.

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Table III - End closure plugs to be furnished with flasks.

Flask type and service	On inlet and outlet end	On drain end
Type GF and CD Service A (non-welded end closure plugs)	Inlet and outlet plug assembly (see figure 10) Inlet and outlet plug details (see figure 1) Inerting cap (see figure 11) Inlet and outlet plug assembly - angle (see figure 13) Inlet and outlet plug - angle details (see figure 4) Figure 13 supplied only when specified (see 6.2)	Drain plug assembly (see figure 10) Drain plug 1/4 inch i.p.s. (see figure 1) Inerting cap 1/4 inch i.p.s.(see figure 12) Internal drain pipe (see figure 14) Figure 14 supplied only when specified (see 6.2)
Type GF and CD Service A (welded end closure plug)	Inlet and outlet plug assembly (see figure 15) Inlet and outlet plug details (see figure 2) Inerting cap (see figure 11) Inlet and outlet plug assembly angle (see figure 16) Inlet and outlet plug - angle details (see figure 4) Figure 16 supplied only when specified (see 6.2)	Drain plug assembly (see figure 15) Drain plug 1/4 inch i.p.s. (see figure 2) Inerting cap 1/4 inch i.p.s. (see figure 12)

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Table III - End closure plugs to be furnished with flasks (cont'd)

Flask type and service	On inlet and outlet end	On drain end
Type GF and CD Service B	Inerting plug (see figure 17) supplied only when no end closure plug is specified (see 6.2)	No drain plug on Service B flasks. One end shall be blank plugged and seal welded (see figure 18) or as specifically approved by the bureau or agency concerned.
Type SF Service A	Inlet and outlet plug (see figure 3) Inerting caps 1 - (see figure 11) 3/8 i.p.s. 1 - (see figure 12) 3/8 i.p.s.	Drain plug assembly (see figure 10) Drain plug 1/4 inch i.p.s. (see figure 1) Inerting cap 1/4 inch i.p.s. (see figure 12)

3.5 Cleaning.-

3.5.1 Flasks for service A shall, prior to painting, be thoroughly cleaned of all mill scale, rust and all other contamination by abrasive blast if necessary, followed by a detergent or other agent not detrimental to the surface to be painted.

3.5.2 Flasks for service B shall be cleaned as specified in 3.5.1 except that the cleaning shall be followed with an alkaline wash with an alkaline solution of not less than 13 pH. having a temperature of 190°F. minimum followed with a hot water rinse of 190°F. minimum. The flask shall be thoroughly dried immediately, subjected to the oil-free test of 4.4.3.5 (all oils that come into contact with a flask under fabrication shall be of a type that will fluoresce under an ultra violet light examination) and inerted as specified in 3.11.

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3.6 Surface protection.- Immediately after cleaning and drying, the flasks shall be given surface protection as follows:

(a) Service A - The interior of service A flasks shall ge given one coat of wash primer, formula 117, in accordance with MIL-P-15328 followed by a number of coats of vinyl zinc chromate, formula 120, in accordance with MIL-P-15930 sufficient to give a total thickness of 3 mils.

(b) Service B - The interior of service B flasks shall not be coated.

(c) Services A and B - The exterior of all flasks shall be coated the same as the interior of service A flasks as specified in 3.6(a).

3.7 Neck threads.- The neck threads shall be Unified National threads, in accordance with Handbook H-28 (see figure 9).

3.8 Pressure.-

3.8.1 Hydrostatic.- Each flask, after final heat treatment, but prior to final cleaning and painting, shall be capable of withstanding the internal hydrostatic test specified in 4.4.3.1.

3.8.2 Air leakage.- Each flask after final assembly shall be capable of withstanding the air leakage test specified in 4.4.3.4.

3.9 Marking for identification.- Permanent marking for identification of flasks shall be stamped on the cylindrical portion of the neck not nearer than 1/2 inch to the tangent at the root of the neck. Such markings shall be made with a round bottom stamp and shall be not over 1/16-inch deep and 1/4 inch in height. Stamping shall be done after final heat treatment and finishing operations, but before painting.

3.9.1 The following information in the order listed shall be stamped on the neck of each flask.

	Example of stamping
(a) Manufacturers initial or trade mark and date of inspection, month and year.	U.S.S., N.T., B & W -- 4-63
(b) Specification, type, class and service.	MIL-F-22606-GF-5000-A

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	Example of stamping
(c) Capacity to nearest tenth cubic foot and weight to nearest pound, and manufacturers serial or code number.	10.0 -- 1705 -- 123-XYZ

3.9.1.1 Identification used by a manufacturer during fabrication, may be stamped longitudinally on the cylindrical portion of the flask prior to forging and heat treating operations. Figures or letters shall not exceed 1/4-inch in height, and shall be made with a round bottom shoulder stamp, 1/16-inch deep. This identification need not be removed after final finishing operations.

3.10 Assembly.- Plugs for non-welded service A flasks shall be assembled with "O" rings and retaining band as shown on figure 10. Plugs shall be torqued to 100 foot pounds. Service A flasks for installation in submarine ballast tanks and service B flasks shall have end plugs torqued to 800 ± 200 foot pounds and seal welded in accordance with figure 10. The completed weld shall be dye penetrant tested in accordance with MIL-STD-271. Welds shall be free of any cracks or porosity.

3.11 Inerting.- Flasks shall be preserved as follows:

(a) Service A - After assembly the flask shall be evacuated to 20 inches of mercury or less. From this partial vacuum, the flask shall be pressurized to 5 to 7 p.s.i.g. with clean, dry air or nitrogen.

(b) Service B - The flask shall be thoroughly dried prior to inerting. The flask shall be evacuated as specified in 3.11(a). From this partial vacuum the flask shall be pressurized to 5 to 7 p.s.i.g. with nitrogen. Commercial nitrogen that is clean, dry and oil-free shall be used.

3.12 Thread coating.- Threads and machined surfaces shall be lubricated with a corrosion inhibiting oxygen resistant compound. The compound shall not deteriorate nor shall it roll, skin, or harden between threads. The compound shall be non-toxic to personnel. The supplier shall certify that the compound will not deteriorate in an atmosphere of pure oxygen when tested in a Parr bomb at 3,000 p.s.i.g. and 200°F. for 72 hours.

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3.13 Workmanship.- Flasks shall be free from cracks. Threads shall be smooth, properly formed and free from burrs, nicks and tears. Paint coatings shall be continuous, adherent and free from blisters streaks, bare spots and other defective conditions indicating poor workmanship.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling.-

4.2.1 Lot.- A lot shall consist of all flasks of the same type, class, service, diameter and wall thickness, from the same heat of steel and heat treated in the same heat treatment furnace charge.

4.2.2 Test sample.- A test sample at least 24 inches long shall be furnished with each lot of flasks. The test sample shall be in the shape of a right circular cylinder having the same diameter and wall thickness as the flasks which it represents. The sample shall be of the same heat of steel and shall be subjected to the same manufacturing processes (except swaging) and heat treatment as the flasks which it represents. The method of quenching the test sample must duplicate that of the flasks. If the flasks are quenched from the outside only, then the test sample must have its ends closed. One end may be vented as necessary to prevent pressure buildup during heat treating. The following tests require tests specimens that shall be cut from the central 12 inch section of the test sample.

(a) Tensile and yield (see 4.4.2.1).
(b) Charpy (see 4.4.2.2) (if Hy-80 steel is used).
(c) Drop weight (NDT) test (see 4.4.2.3) (if A336 steel is used).

4.3 Visual and dimensional examination.- Each flask shall be visually and dimensionally examined to verify compliance with this specification. Flasks shall be clean and free from loose scale, chips, or foreign materials. Flasks containing visual or dimensional defects shall not be offered for delivery. Surface imperfections such as round bottom pits, scores, gouges and similar defects are not cause for rejection unless the depth of the imperfection violates the minimum design wall thickness of the flask. Such imperfections if greater than 1/16-inch deep shall be faired at not less than a 4 to 1 ratio into the adjacent area by grinding or machining.

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4.4 Test procedures.-

4.4.1 Chemical analysis.- Prior to flask fabrication a suitable sample shall be taken from the pipe, billets or plate of each heat of steel, to be used for flask fabrication, and shall be subjected to a chemical analysis to determine compliance with 3.1.1. Sample selection and testing shall conform to test method 111.1 of FED-STD-151. Failure of the chemical analysis to conform as specified shall be cause for rejection of the heat of material represented.

4.4.2 Destructive tests.- The tests in this section require test specimens cut from the central 12 inch section of the test sample (see 4.2.2).

4.4.2.1 Tensile and yield.- Four tensile specimens shall be cut longitudinally from the specified portion of the test sample (see 4.2.2) spaced 90 degrees apart and shall be tested in accordance with method 211.1 of FED-STD-151 to determine compliance of tensile, yield and elongation properties specified in 3.2. Failure to pass this test shall be cause for rejection of the lot represented.

4.4.2.2 Charpy test (applicable only if flask is fabricated from Hy-80 steel).- Three Charpy V-notch test specimens shall be cut from the specified portion of the test sample (see 4.2.2) and tested in accordance with method 221.1 of FED-STD-151 to determine compliance with 3.2. The specimens shall be selected in accordance with the instructions for plate. The value specified in 3.2 shall be the average of three specimens. However, no single specimen shall be below the specified average by more than 5 foot pounds. Failure to pass this test shall be cause for rejection of the lot represented.

4.4.2.3 Drop weight test.- The drop weight test shall be conducted in accordance with NAVSHIPS 250-634-3. Not less than three specimens taken from each test sample (see 4.2.2), shall be tested at a temperature of 10°F. No specimen shall fail. This indicates an NDT temperature of zero degrees in accordance with the method defined in NAVSHIPS-250-634-3. Where the flask wall thickness precludes the use of a 5 inches by 2 inches by 5/8 inch test specimen, a 5 inches by 2 inches by 1/2 inch test specimen may be utilized. The dimension for the deflection stop for the 1/2 inch thick specimen shall be 0.090 inch, plus 0.004 inch, minus 0 inch in lieu of 0.075 inch + .004.
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4.4.3 Flask tests.-

4.4.3.1 Hydrostatic test.- Each flask shall be tested in a water jacket to a test pressure equal to 5/3 the design working pressure of the flask. Cold tap water shall be used for this test. Neither the water nor the flask shall be heated. The test pressure shall be held at least 30 minutes. As a result of this test the flask shall not rupture or show a permanent set exceeding 10 percent of the total volumetric expansion. Any flask that fails this test shall not be offered for delivery.

4.4.3.2 Inspection.- Hemispherical ends and necks of flasks shall be inspected by the magnetic particle method in accordance with MIL-STD-271 after the hydrostatic test has been completed.

4.4.3.3 Ultrasonic inspection.- After the hydrostatic test has been completed, each flask shall be 100 percent inspected over the cylindrical section either in a circumferential direction or a longitudinal direction between the heads in accordance with one of the methods prescribed in MIL-STD-271. The equipment shall be calibrated to detect a notch equal to 3 percent of the wall thickness.

4.4.3.3.1 Shear wave testing acceptance criteria.- Any flask having a discontinuity indication greater than that produced by the 3 percent calibration notch shall not be offered for delivery. Any flask having a discontinuity equivalent to 50 percent of that produced by the calibration notch, and over 3 inches in length, shall be rejected unless approved by the bureau or agency concerned. Other ultrasonic indications over 25 percent of that produced by the calibration notch, shall also be submitted to the bureau or agency concerned for information.

4.4.3.3.2 Through wave testing acceptance criteria.- For the through wave testing method, a 100 percent loss of back reflection indicates a discontinuity size equal to or greater than the 3 percent calibration notch. Any flask showing such a discontinuity size shall not be offered for delivery. All flasks having discontinuities producing a loss in back reflection between 50 percent and 100 percent that are greater than 3 inches in length shall be reported to the bureau or agency concerned together with all other discontinuities producing a loss in back reflection greater than 25 percent.

4.4.3.3.3 Ultrasonic inspection equipment shall be recalibrated periodically to insure optimum results. It shall also be recalibrated if there is any change in frequency or crystal size used.

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4.4.3.3.4 Reports.- A report shall be submitted in sketch form orientating all reportable discontinuities longitudinally or circumferentially as found, and indicating their severity and length. If the reported discontinuities are lineally disposed along the flask and the total length of all discontinuities in 12 inches of flask length is 6-inches, the flask shall not be offered for delivery.

4.4.3.4 Air leakage test.- After the hydrostatic test of 4.4.3.1 and inspection specified in 4.4.3.2 and 4.4.3.3 each flask shall be pressurized with air to the design working pressure of the flask. Test pressure shall be held for not less than 5 minutes. There shall be no evidence of leakage through any portion of the flask or end plug. Any flask that fails to comply with this test shall not be offered for delivery.

4.4.3.5 Oil-free test.- Service B flasks, after cleaning as specified in 3.5.2, shall be inspected internally by ultra violet light ("Black light": 3200 to 4000 Angstrom units). If there is any evidence of fluorescence on the interior or machined surfaces the flask shall be recleaned (see 3.5.2) and resubmitted to the oil-free test.

4.4.3.6 Retests.-

4.4.3.6.1 If specimens fail tests specified in 4.4.2.1, 4.4.2.2 and 4.4.2.3, the manufacturer will be permitted to reheat treat the lot, and repeat the tests specified in 4.4, except chemical analysis (see 4.4.1), air leakage (see 4.4.3.4) and oil-free test (see 4.4.3.5).

4.4.3.6.2 Surface imperfections detected by 4.4.3.2 or 4.4.3.3 which would require rejection but can be made acceptable by grinding or machining, may be so treated and retested.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation.-

5.1.1 Packaging.- Flask inerted with nitrogen (see 3.11) shall be packaged in a manner sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

5.1.1.2 Packing.- Packing shall be accomplished in a manner which will insure acceptance by common carrier and will afford protection against physical and mechanical damage from handling during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation.

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5.1.1.3 Marking.- Shipment marking information shall be provided in accordance with the supplier's commercial practice. The information shall include nomenclature, contract or order number, manufacturer's name and destination. Securely attached to each flask shall be a waterproof tag indelibly marked with all pertinent shipping information.

5.2 Domestic shipment and storage or overseas shipment.- The requirements and levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.2).

(5.2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made.

5.2.1.1 Packaging.- Not applicable.

5.2.1.2 Packing.- Flasks, filled with nitrogen (see 3.11), of like type shall be packed for shipment as specified in 5.2.1.2.1.

5.2.1.2.1 Levels A and B.- Type GF and type SF flasks shall be pallitized for shipment in accordance with MIL-STD-147. Type CD flask (curved design) shall be packed in open wood crates conforming to MIL-C-132, style B, with the exception of limitation on container size. The crate closure and strapping and the anchoring of contents shall be in accordance with the appendix to MIL-C-132. Shrouds shall not be required.

5.2.1.3 Marking.- In addition to any special marking required in the contract or order, shipments and shipping containers shall be marked in accordance with MIL-STD-129 and the appendix to MIL-C-132).

6. NOTES

6.1 Intended use.- These flasks are intended for but not limited to shipboard use in the storage of high-pressure air, oxygen and nitrogen.

6.2 Ordering data.- Procurement documents shall specify the following:

(a) Title, number and date of this specification.
(b) Type, class and service required (see 1.2).
(c) Capacity of flasks (see 3.3).
(d) End plug design and size and type of connection (see table III).
(e) If internal drain plugs are required (see table III).
(f) Preservation, packaging, packing or marking requirements if other than those specified in 5.1).

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6.3 Procuring activities should include the following as part of the invitation for bid (see 3.4).

"Flasks of welded construction will not be considered under this procurement request unless the welding and fabrication procedures and operator qualification have been qualified in accordance with the requirements of appendix I and approved by the Bureau of Ships."

6.4 Selection instructions.- For standardization and stocking purposes the following will be used as a guide for procuring high pressure flasks:

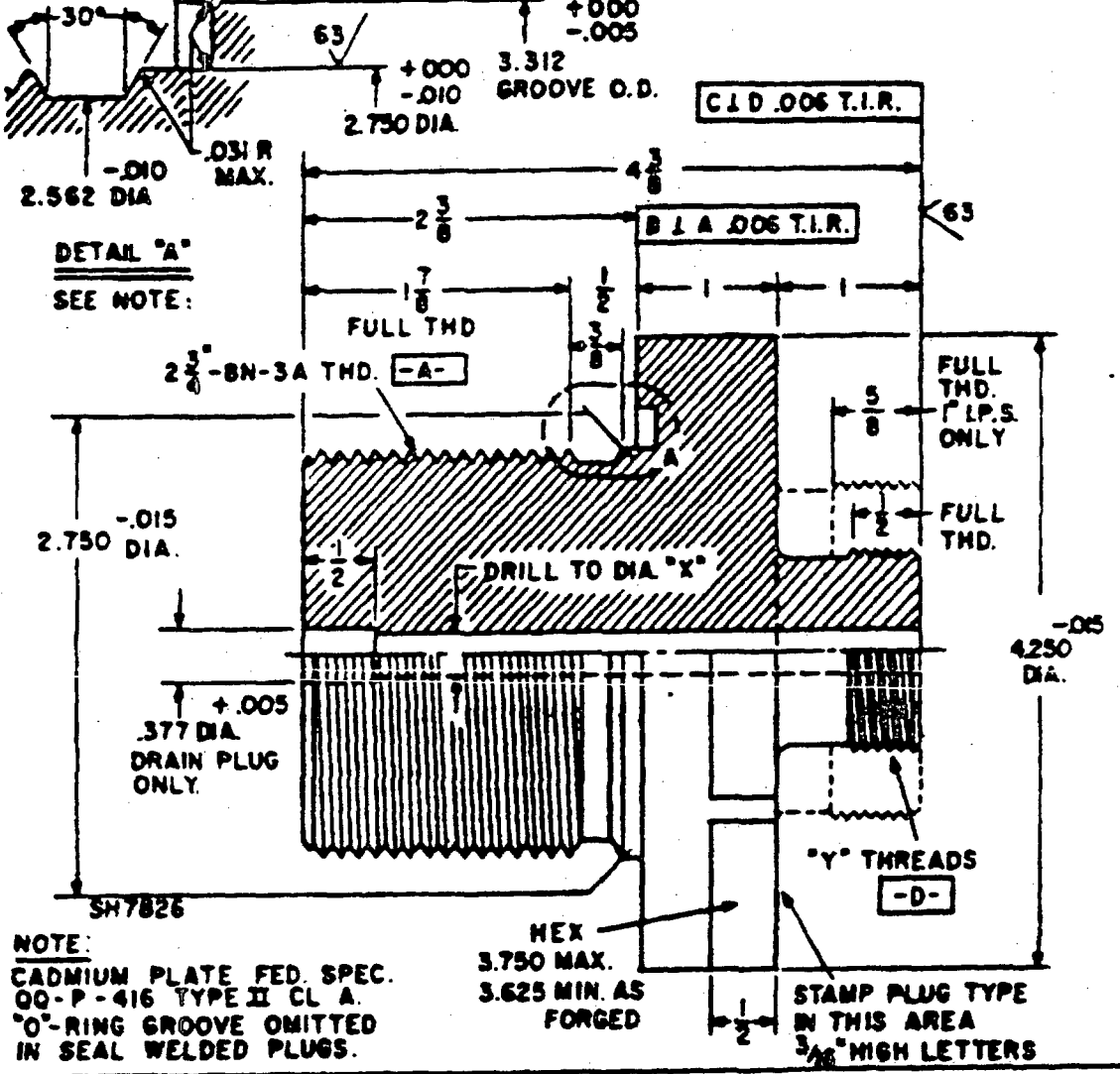
(a) Flasks will be selected from the cubic capacities and working pressures listed in table II. When the desired capacity falls between the standard capacities listed, the next higher capacity shall be used. When the designed working pressure falls between those listed, the next higher working pressure shall be used. Where it is impractical to conform to the standard capacities or working pressures listed in table II, specific approval of the cognizant activity must be obtained prior to procuring any special flasks.

(b) The 1/2 cubic foot type SF separator and 1-1/2 cubic foot GF flask will be procured in the 5000 p.s.i. class only and shall be used for all pressures up to and including 5000 p.s.i. When a separator with a larger cubic capacity is needed, consideration should be given to the 1-1/2 cubic foot flask.

6.5 CHANGES FROM PREVIOUS ISSUE. THE EXTENT OF CHANGES (DELETIONS, ADDITIONS, ETC.) PRECLUDE THE ANNOTATION OF THE INDIVIDUAL CHANGES FROM THE PREVIOUS ISSUE OF THIS DOCUMENT.

Preparing activity:
Navy - Ships
(Project 6120-N020Sh)

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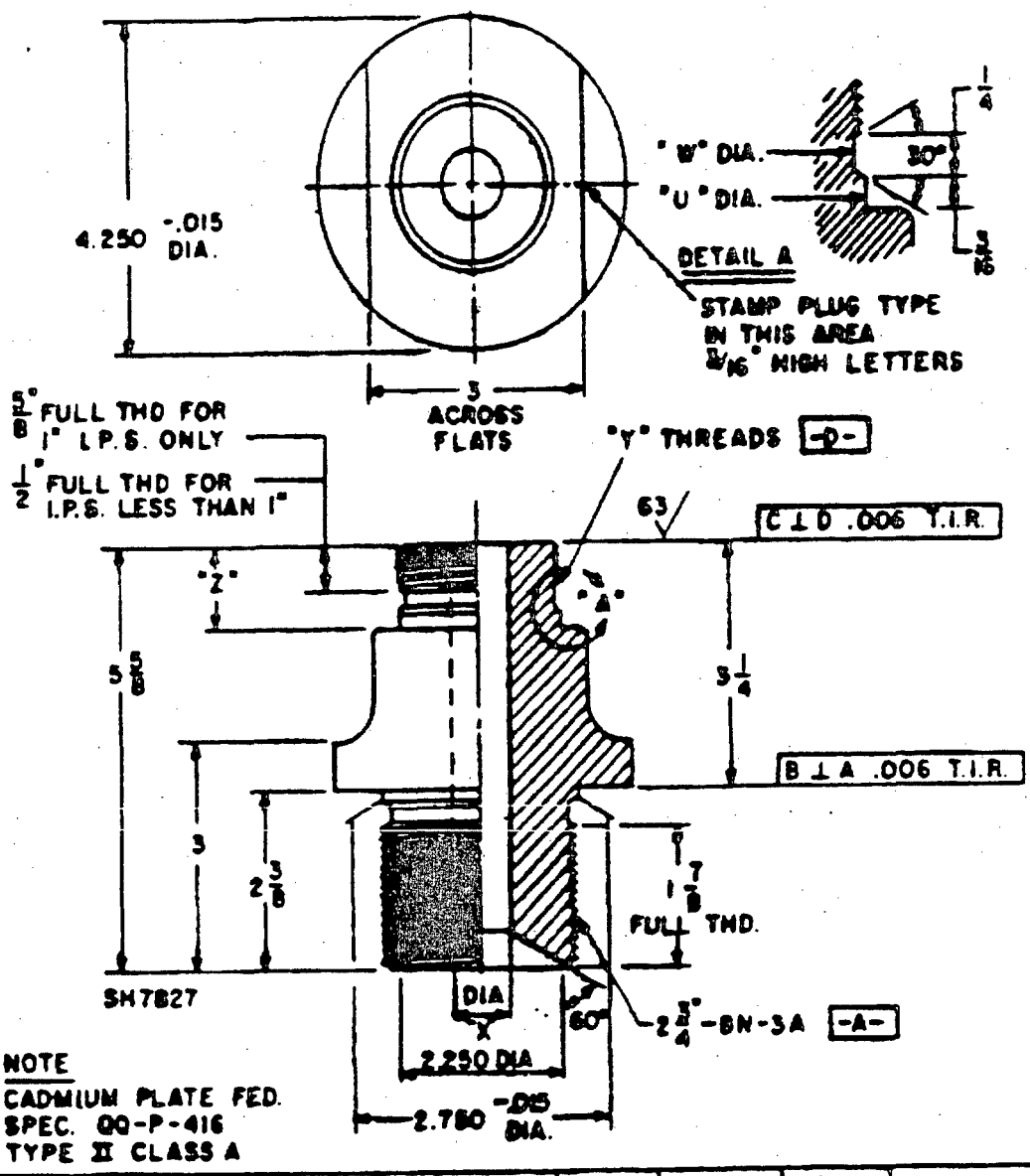


NOTE:
CADMIUM PLATE FED. SPEC. QQ-P-416 TYPE II CL A.
"O"-RING GROOVE OMITTED IN SEAL WELDED PLUGS.

IRON PIPE SIZE	X DIA.	Y THREADS	REMARKS
1/4"	5/16"	1 3/16"-12UN-3A	DRAIN PLUG
3/8"	7/16"	1 3/8"-12UNF-3A	INLET AND OUTLET PLUG
1/2"	1/2"	1 3/4"-12 UN-3A	INLET AND OUTLET PLUG
3/4"	5/8"	2"-12UN-3A	INLET AND OUTLET PLUG
1"	13/16"	2 5/16"-12UNS-3A	INLET AND OUTLET PLUG
SOLID	-	-	END CLOSURE PLUG-NO BOSS REQ'D

FIGURE 1 - PLUGS FOR "O" RING ASSEMBLY

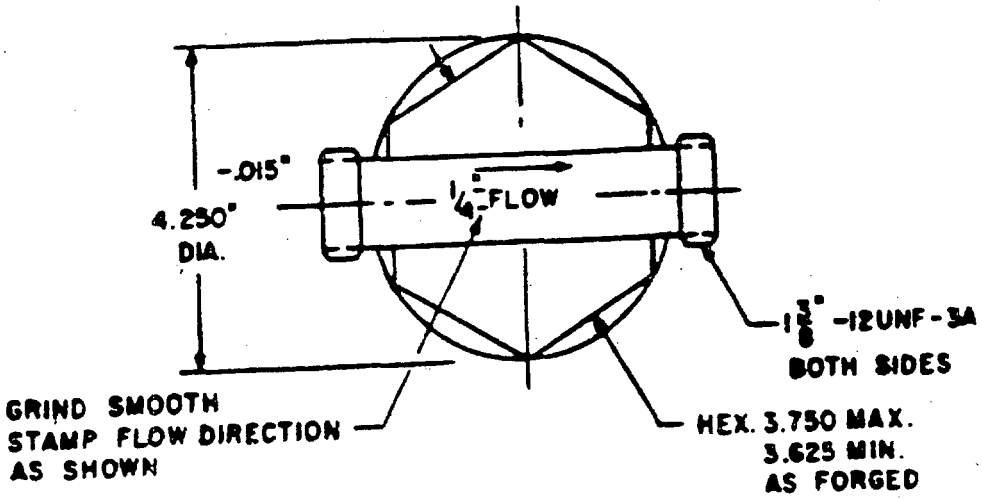
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IRON PIPE SIZE	"X" DIA	"Y" THREADS	"Z" ±.006	"U" DIA. -.010	"W" DIA -.015	
1/4"	5/16"	1 3/16"-12 UN-3A	.916	1.187	1.075	
3/8"	7/16"	1 3/8"-12 UNF-3A	.916	1.375	1.262	
1/2"	1/2"	1 3/4"-12 UN-3A	.916	1.750	1.637	
3/4"	5/8"	2"-12 UN-3A	.916	2.000	1.887	
1"	13/16"	2 5/16"-12 UNS-3A	1.041	2.312	2.200	
1 1/2"	1 5/16"	1 3/4"-12 UN-3	.916	1.750	1.637	SPECIAL

FIGURE 2 INLET/OUTLET PLUGS FOR WELDED ASSEMBLY

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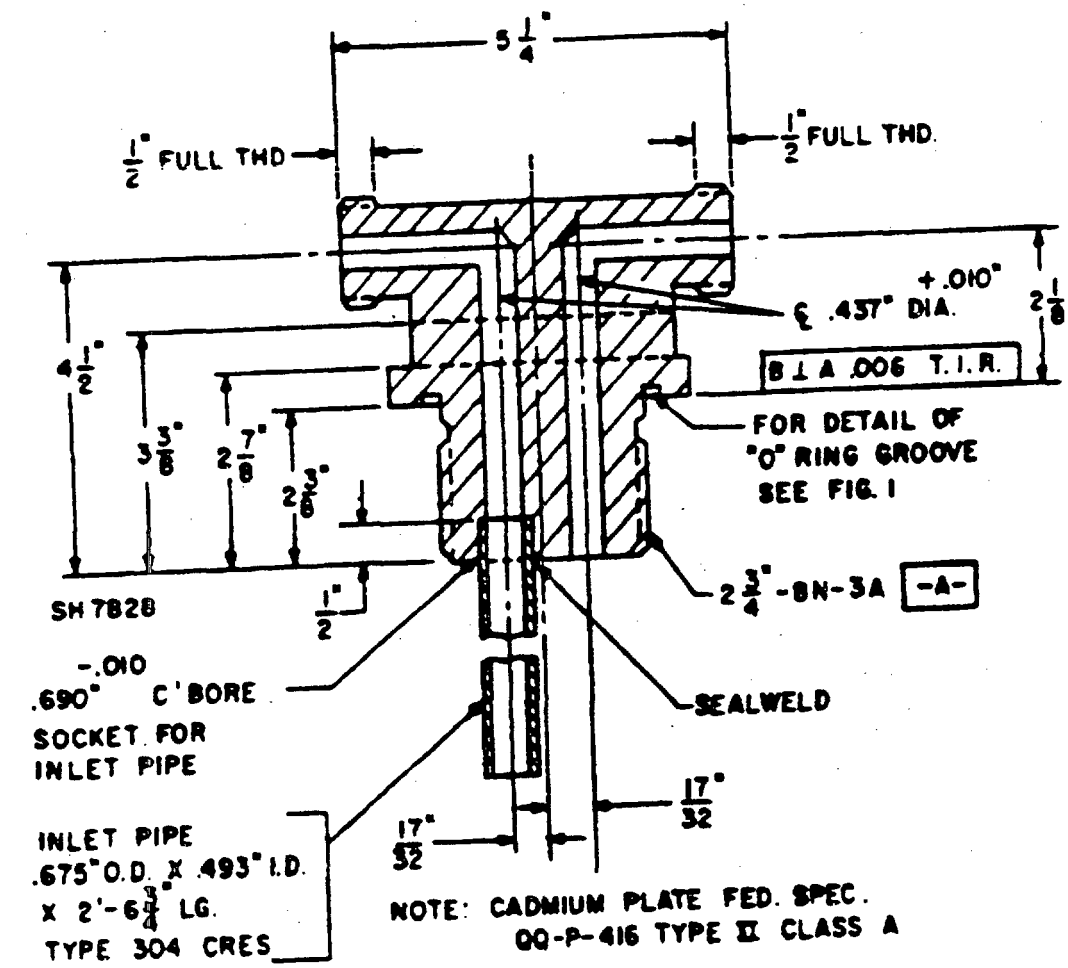
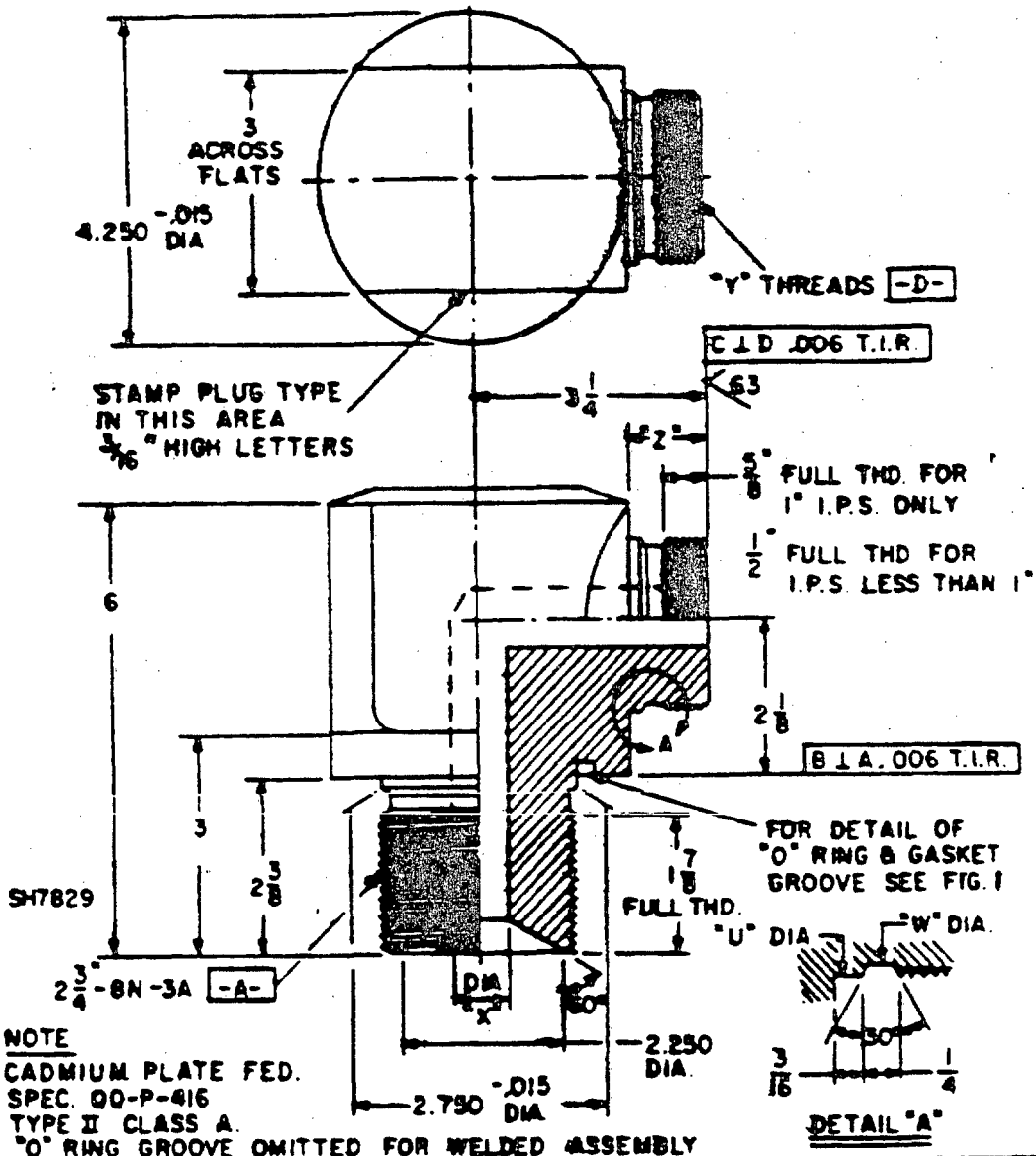


FIGURE 3 INLET AND OUTLET PLUG
FOR TYPE SF (SEPARATOR) FLASK

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NOTE
CADMIUM PLATE FED. SPEC. QQ-P-416 TYPE II CLASS A.
"O" RING GROOVE OMITTED FOR WELDED ASSEMBLY

IRON PIPE SIZE	"X" DIA.	"Y" THREADS	"Z" +.006	"U" DIA -.010	"W" DIA -.015	
1/4"	5/16"	1 3/16"-12UN-3A	.916	1.187	1.075	
3/8"	7/16"	1 3/8"-12UNF-3A	.916	1.375	1.262	
1/2"	1/2"	1 3/4"-12UN-3A	.916	1.750	1.637	
3/4"	5/8"	2"-12UN-3A	.916	2.000	1.887	
1"	13/16"	2 5/16"-12UNS-3A	1.041	2.3125	2.200	
1/2"	13/16"	1 3/4"-12UN-3A	.916	1.750	1.637	SPECIAL

FIGURE 4 INLET/OUTLET PLUGS ANGLE TYPE FOR WELDED OR "O" RING ASSEMBLY

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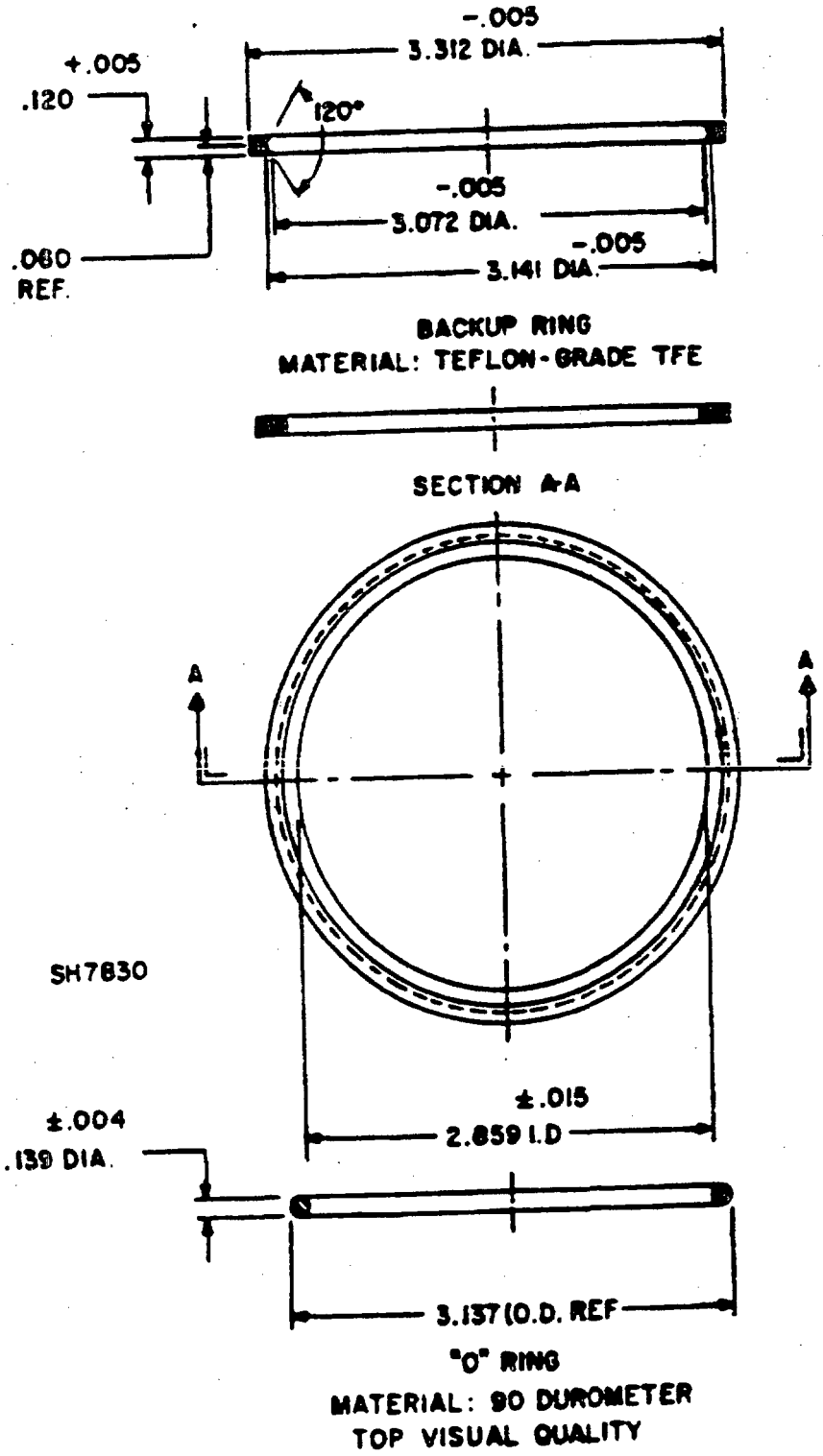


FIGURE 5 "O" RING AND BACKUP RING ASSEMBLY

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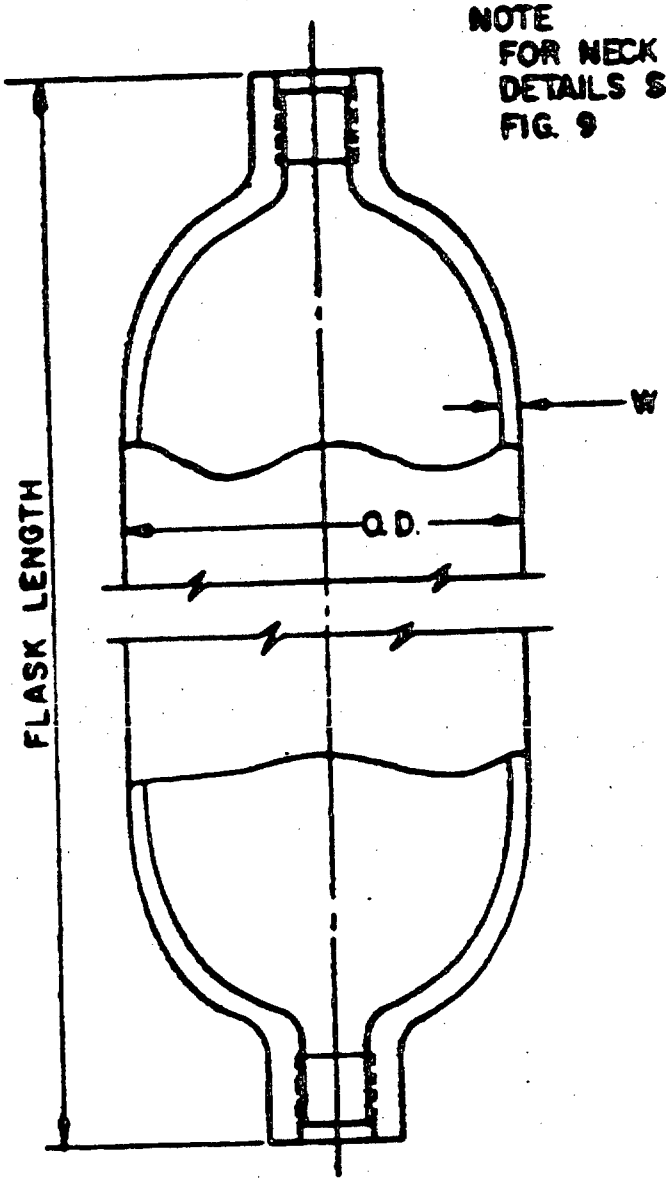
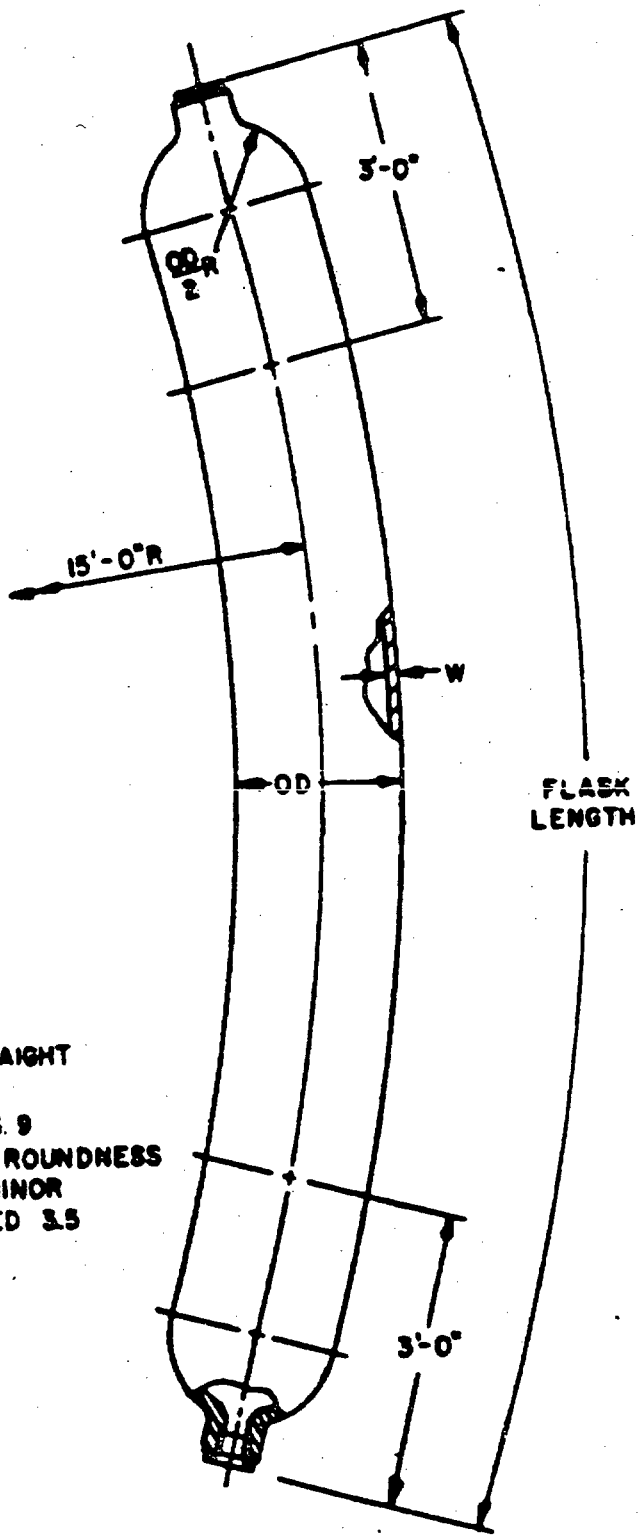


FIGURE 6 TYPE GF FLASK

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BOTH ENDS ALIKE



NOTES:
1. 36" END SECTION MAY BE STRAIGHT TO FACILITATE FABRICATION.
2. FOR NECK DETAILS SEE FIG. 9
3. THE DIFFERENCE IN OUT OF ROUNDNESS BETWEEN THE MAJOR AND MINOR DIAMETER SHALL NOT EXCEED 3.5 PERCENT OF THE OD.

SH7632

FIGURE 7 TYPE CD FLASK

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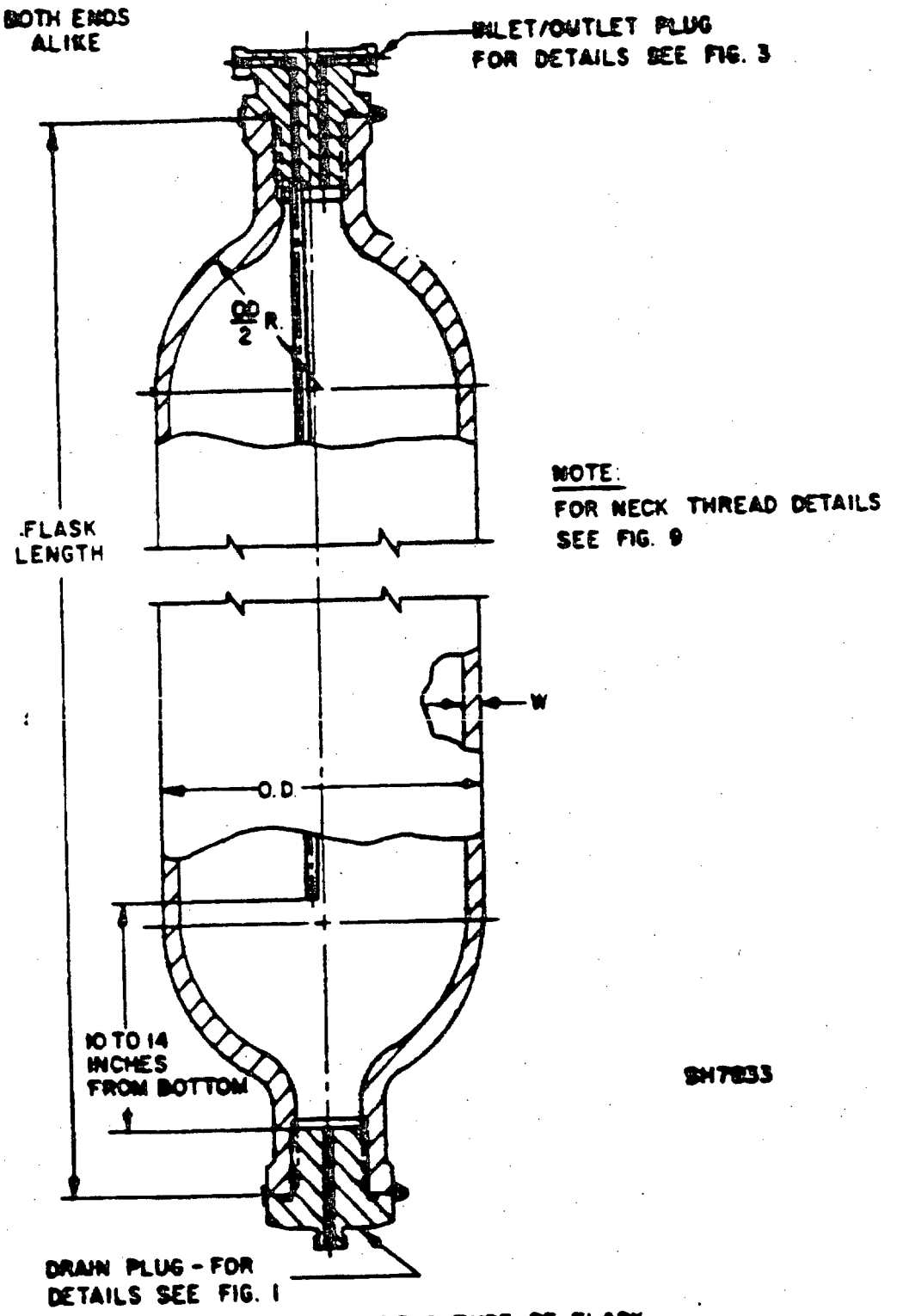
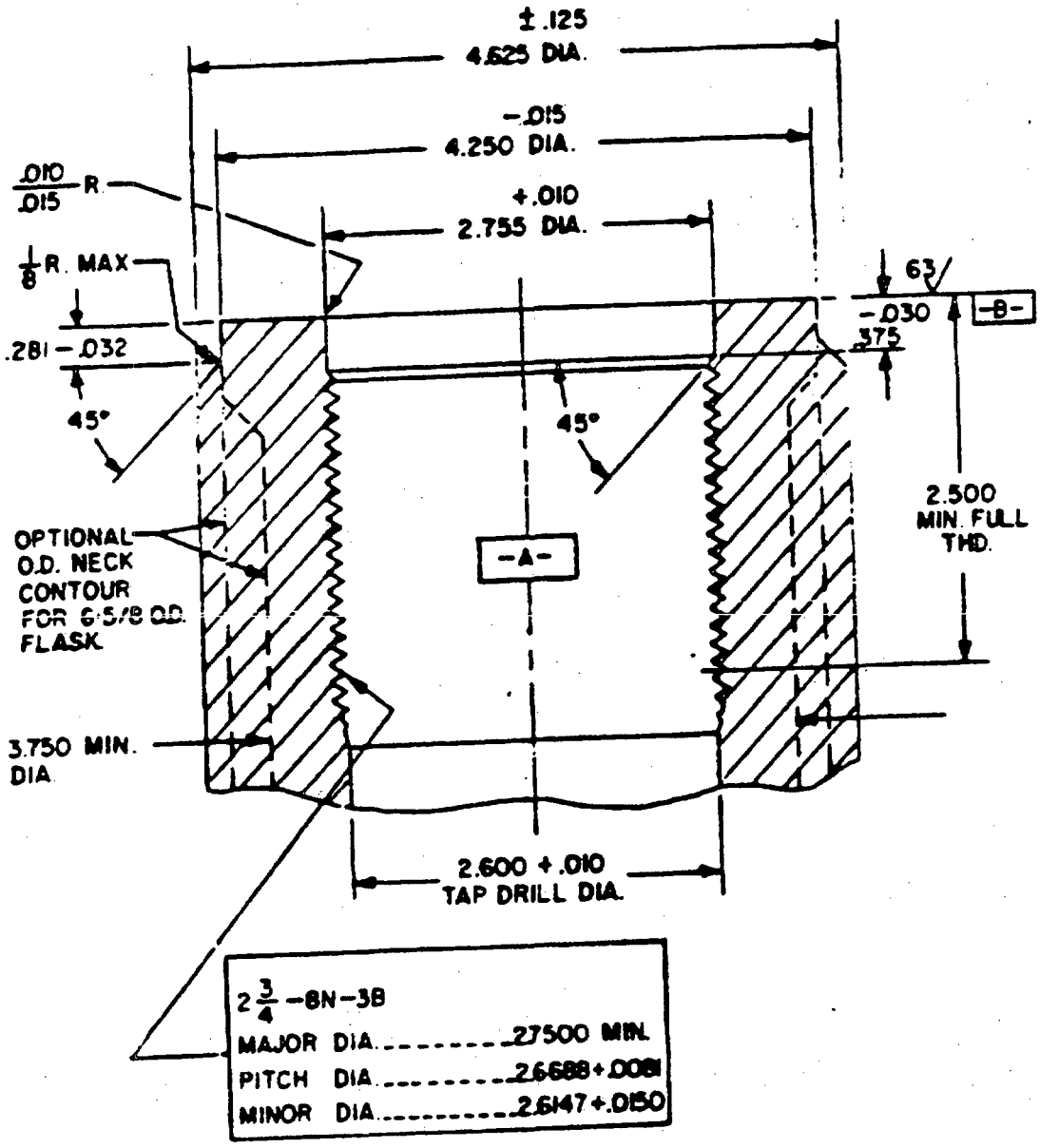


FIGURE 8 TYPE SF FLASK

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NOTE: SURFACE -B- MUST BE PERPENDICULAR TO AXIS -A- WITHIN .006 T.I.R.

SH 7834

FIGURE 9 NECK THREAD DETAILS

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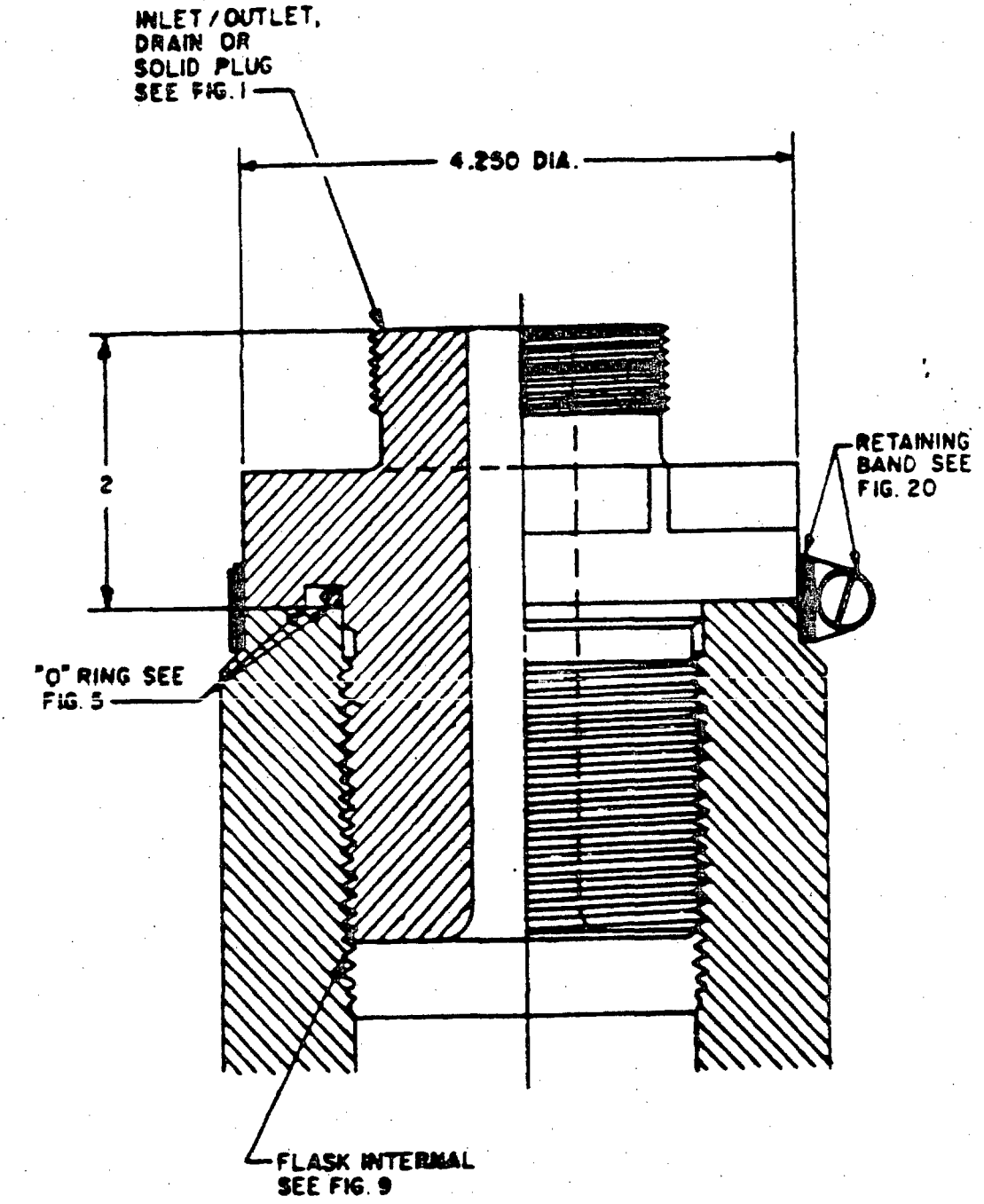
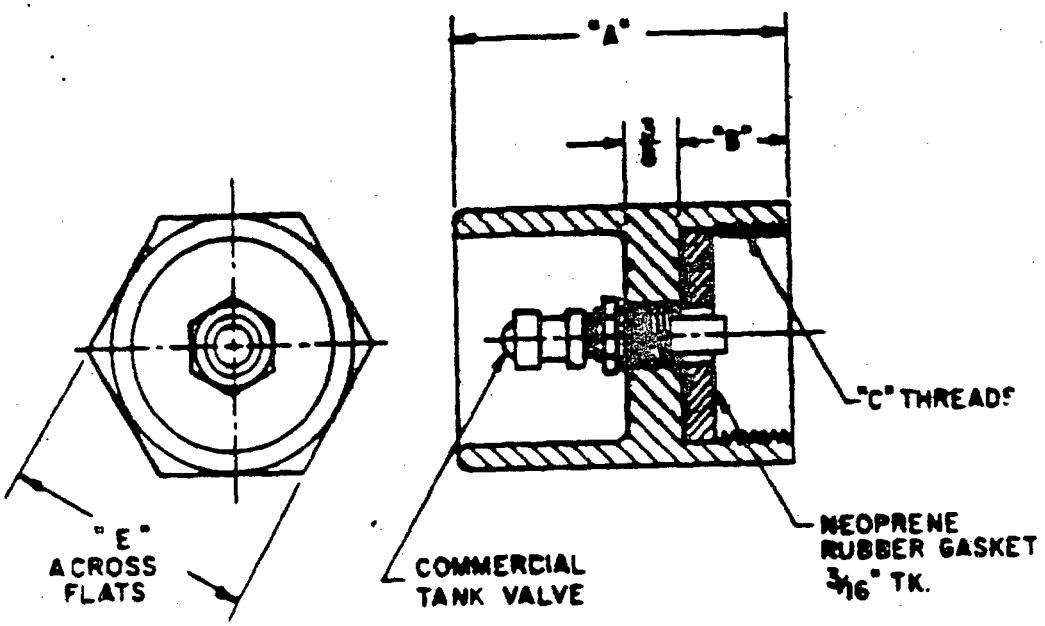


FIGURE 10 INLET/OUTLET PLUG & "O" RING NONWELDED ASSEMBLY

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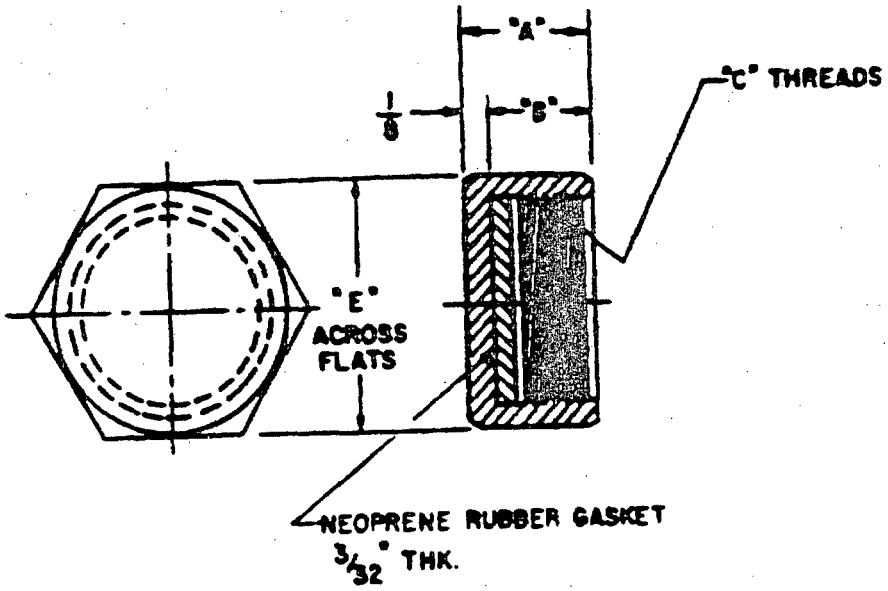
SH 7836

NOTE:
CADMIUM PLATE FED. SPEC.
QQ-P-416 TYPE II CLASS A

IRON PIPE SIZE	"A" OVERALL	"B"	"C" THREADS	"D" GASKET O.D.	"E" HEX.
1/4"	2 1/8"	5/8"	1 3/16" - 12 UN - 3B	1 3/16" MAX.	1 3/8"
3/8"	2 1/4"	3/4"	1 3/8" 12 UNF - 3B	1 3/8" "	1 5/8"
1/2"	2 1/4"	3/4"	1 3/4" - 12 UN - 3B	1 3/4" "	2"
3/4"	2 1/4"	3/4"	2" - 12 UN - 3B	2" "	2 1/4"
1"	2 3/8"	7/8"	2 5/16" - 12 UNS - 3B	2 5/16" "	2 1/2"

FIGURE 11-INERTING CAP WITH TANK VALVE

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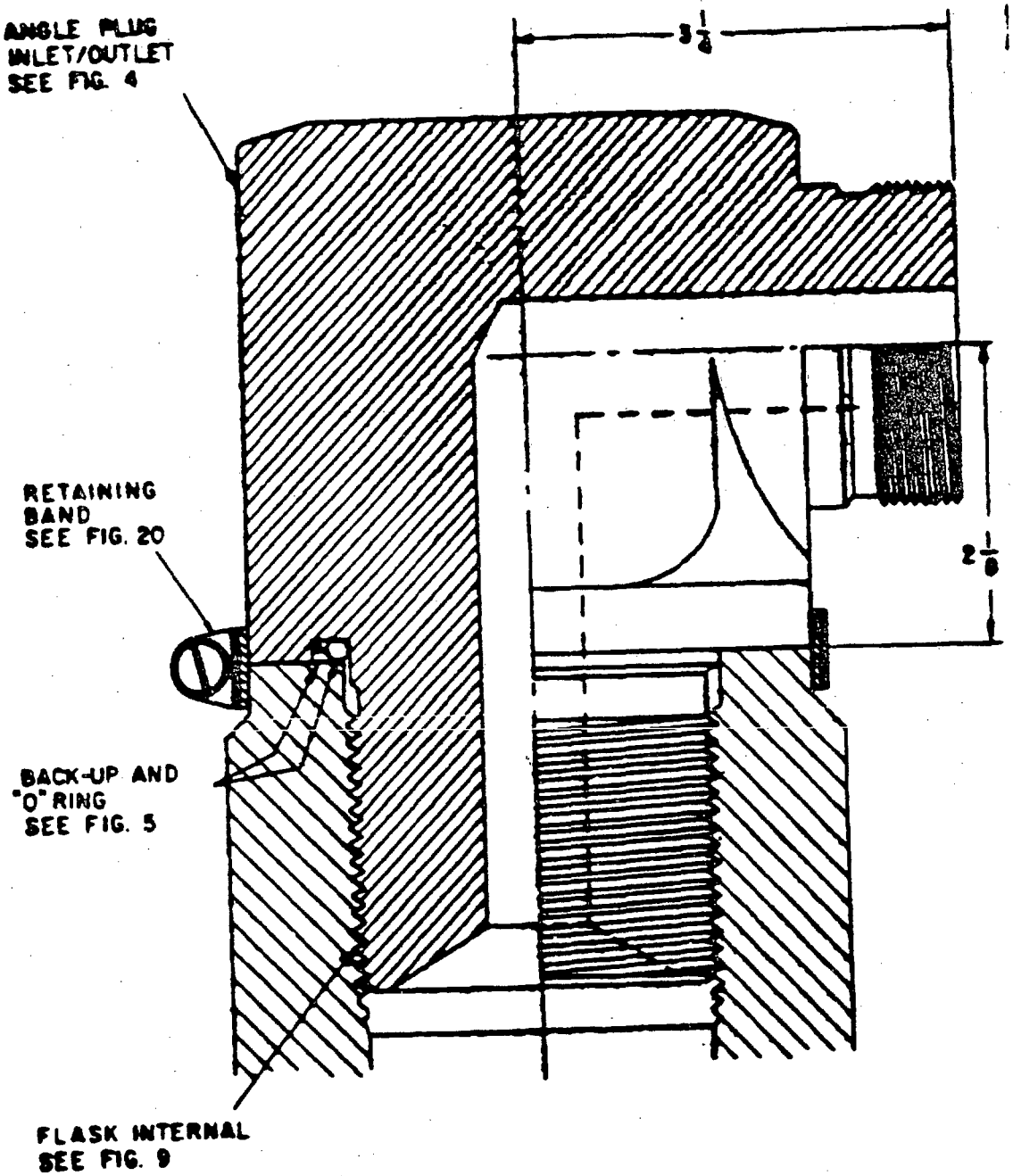
SH7837

NOTE:
CADMIUM PLATE FED. SPEC. QQ-P-416
TYPE-II-CLASS A

IRON PIPE SIZE	"A" OVERALL	"B"	"C" THREADS	"D" GASKET O.D.	"E" HEX.
1/4"	3/4"	5/8"	1 3/16" -12UN-2B	1 3/16"	1 3/8"
3/8"	7/8"	3/4"	1 3/8" - 12 UNF-2B	1 3/8"	1 5/8"
1/2"	7/8"	3/4"	1 3/4" - 12 UN-2B	1 3/4"	2"
3/4"	7/8"	3/4"	2" - 12 UN-2B	2"	2 1/4"
1"	1"	7/8"	2 5/16" -12 UNS-2B	2 5/16"	2 1/2"

FIGURE 12 INERTING CAP, SOLID

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SH7838

FIGURE 13 ANGLE INLET/OUTLET PLUG & "O" RING ASSEMBLY

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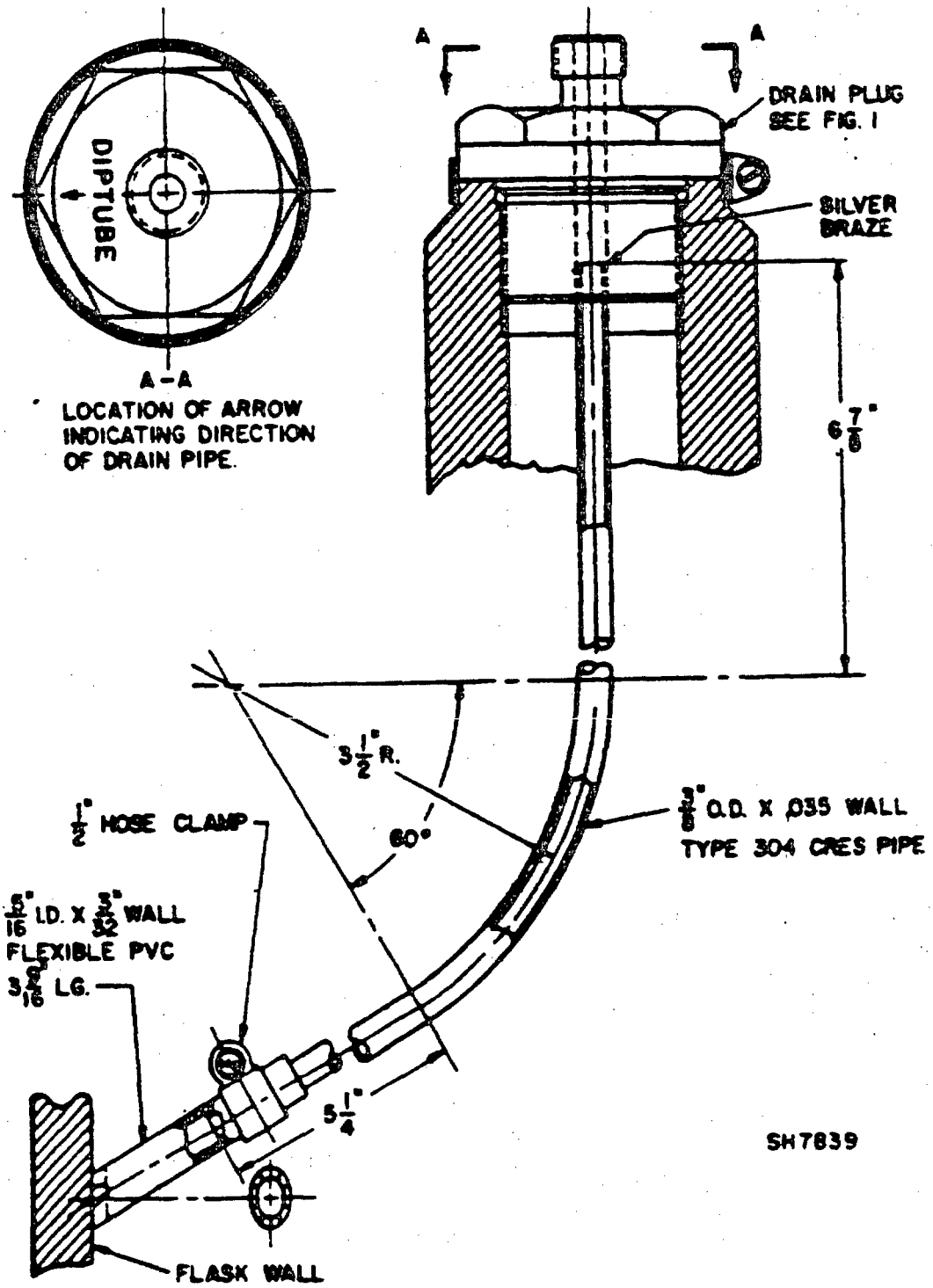
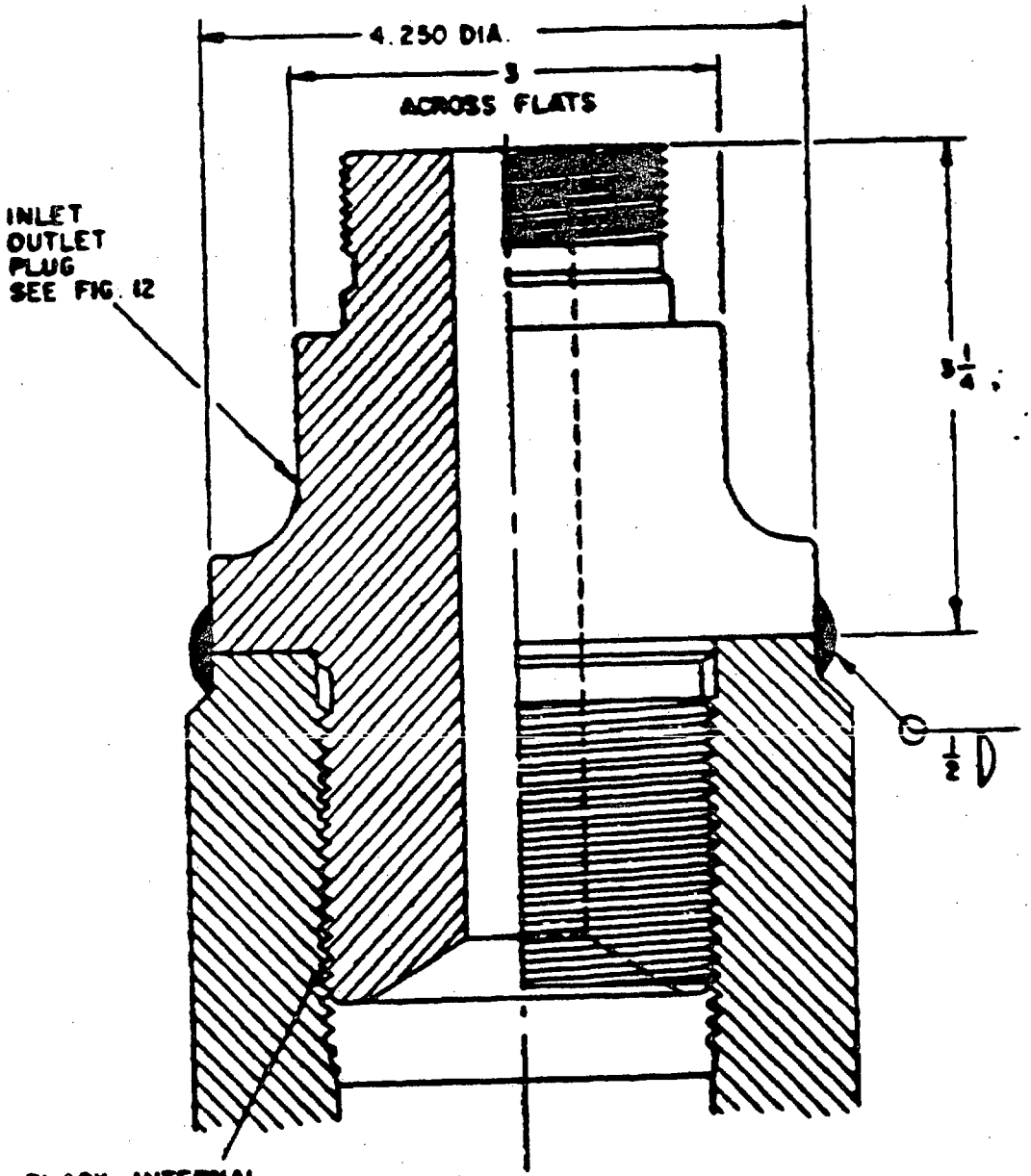


FIGURE 14 INTERNAL DRAIN PIPING ASSEMBLY FOR FLASKS INSTALLED HORIZONTALLY

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SH7840

FIGURE 15 TYPICAL INLET/OUTLET PLUG SEAL WELDED ASSEMBLY

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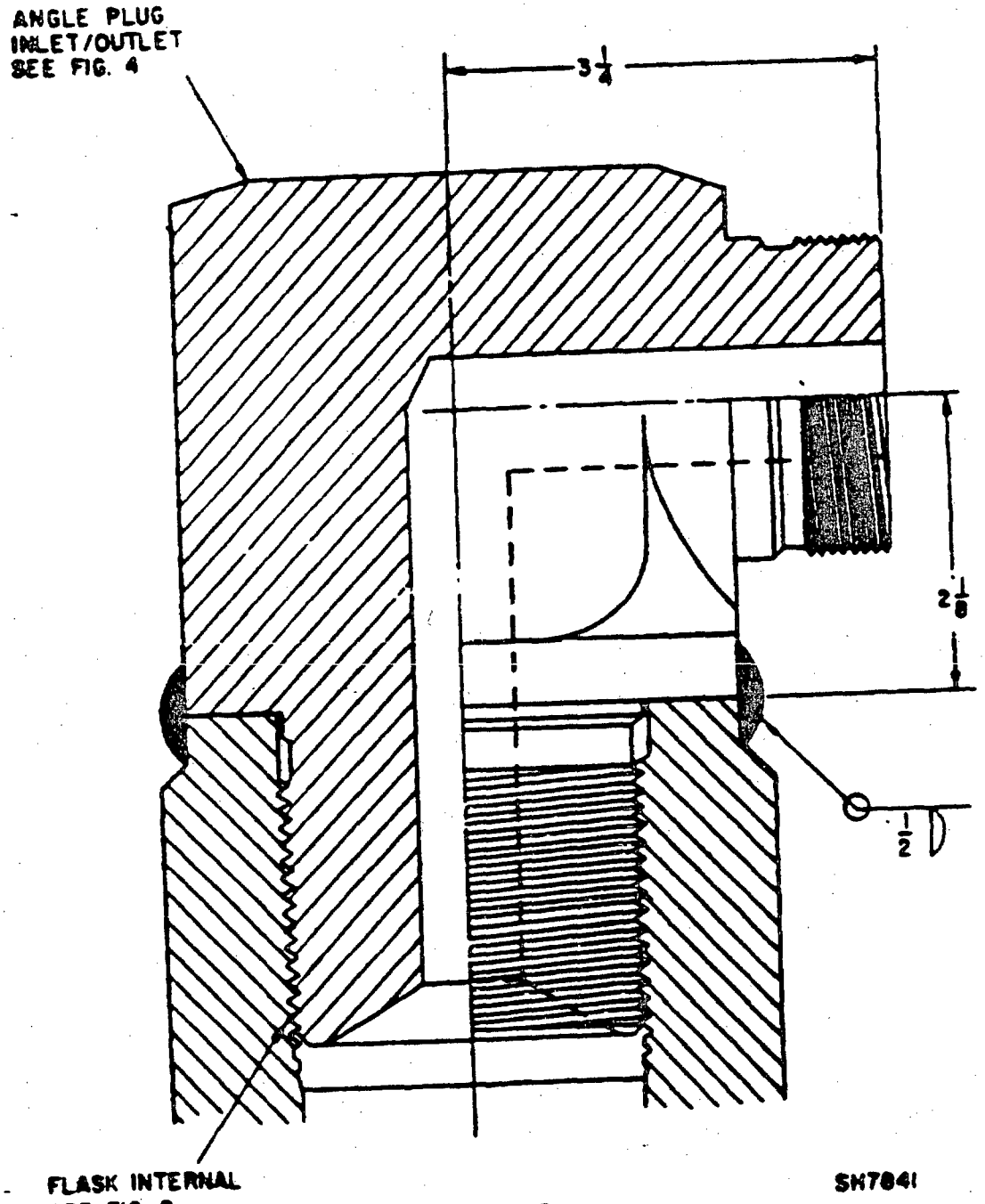
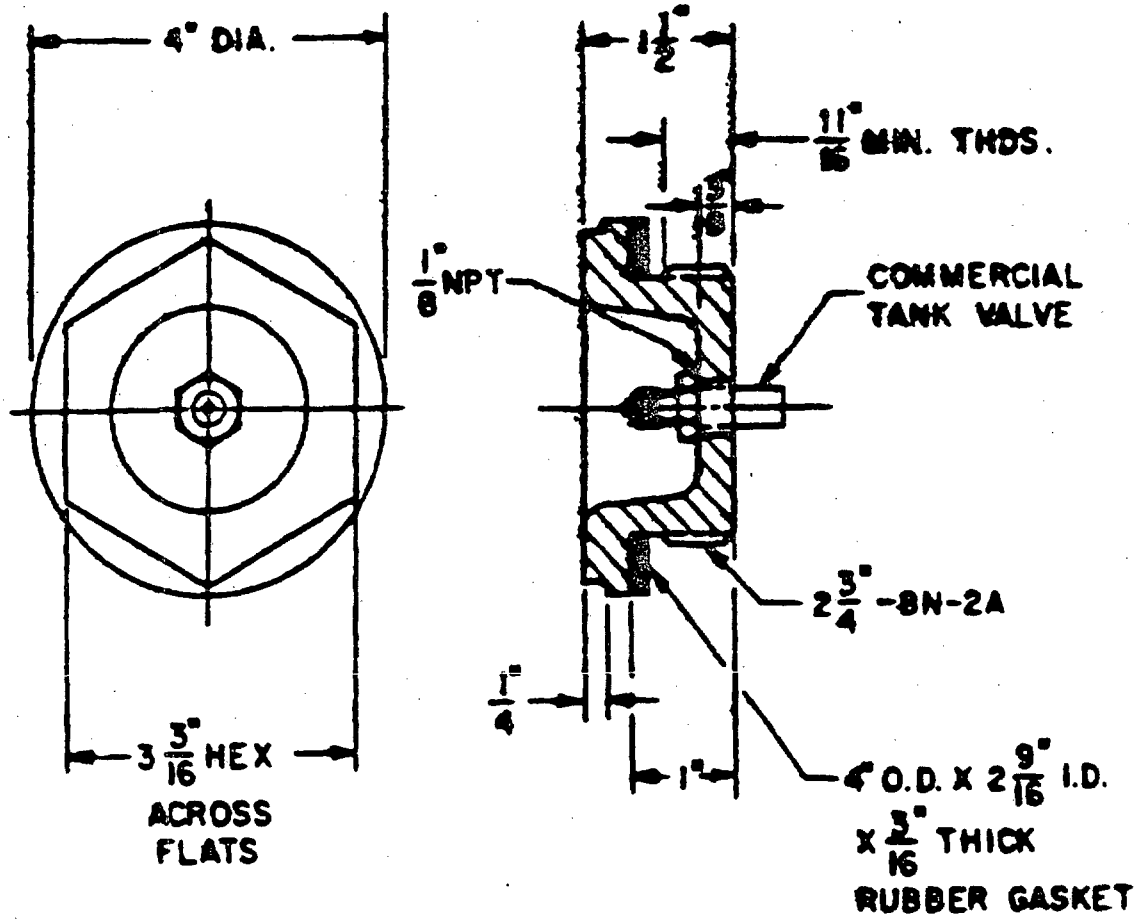


FIGURE 16
TYPICAL ANGLE PLUG WELDED ASSEMBLY

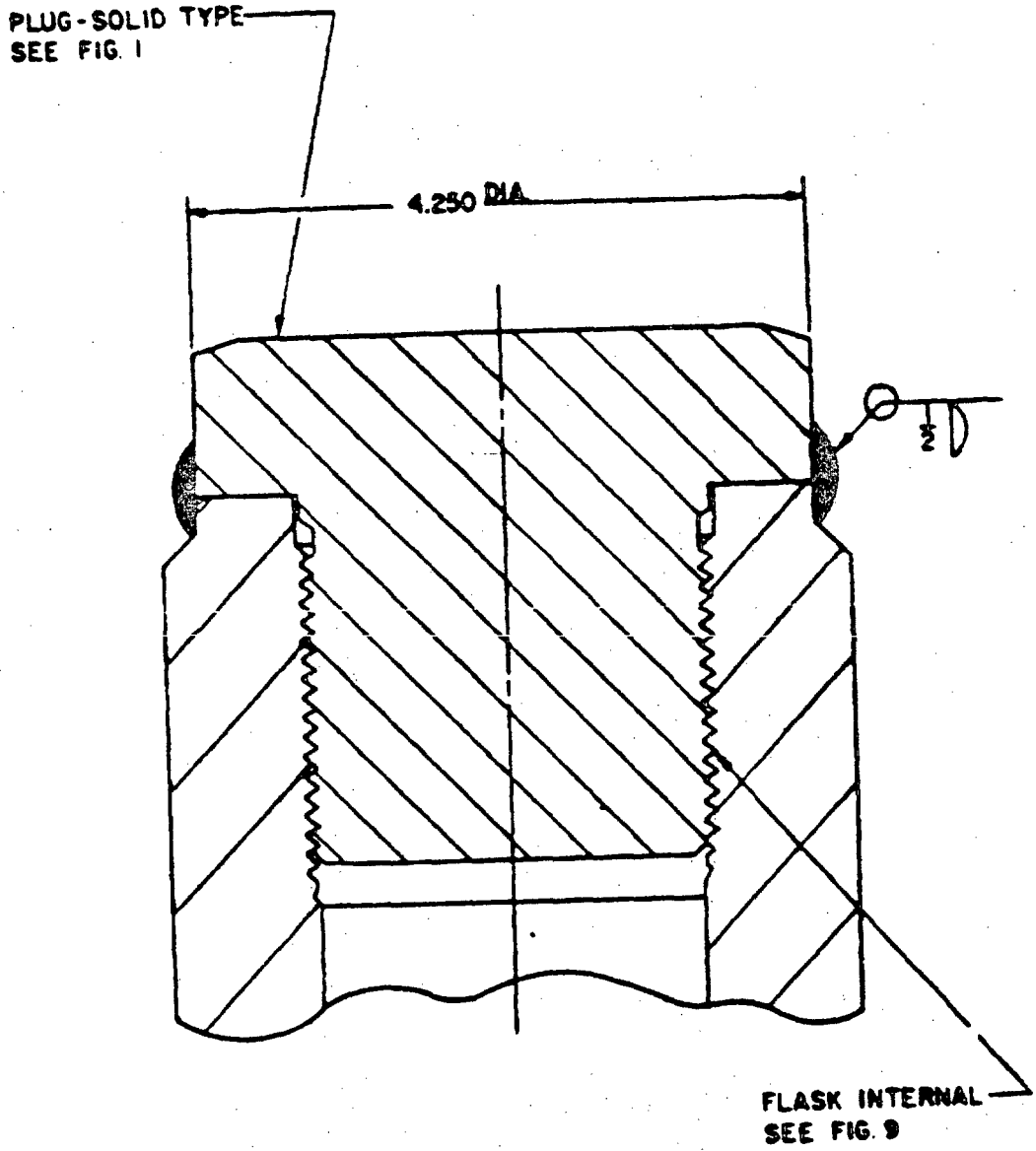
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SH7842

FIGURE 17 INERTING PLUG

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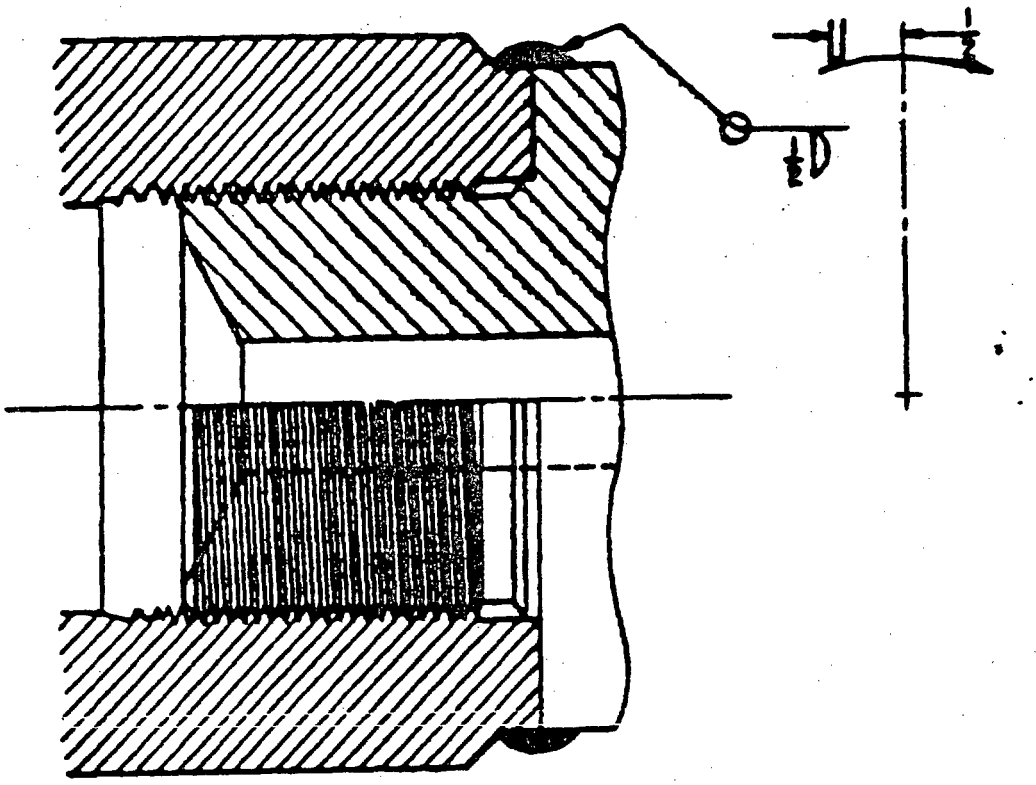


SH7843

FIGURE 18 DETAIL OF PLUG ON SEALED END

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TYPE OF WELD-SEAL WELD



SHT844

SEAL WELD PROCEDURE

TORQUE PLUG TO 800 ± 200 FOOT POUNDS

PREPARATION - CLEAN AREA TO BE SEAL WELDED INCLUDING ABUTTING FACES OF CYLINDER NECK AND PLUG.

POSITION - HORIZONTAL ROLL WITH ELECTRODE 1/2" AHEAD OF TOP CENTER AS ILLUSTRATED.

PREHEAT - 100° F. TO 200° F. MAX.

ELECTRODE - AWS ASTM E 309-16 PREFERRED, 1/8" DIAMETER, USING D.C. REVERSE POLARITY CURRENT AT 24-26 VOLTS AND 80-100 AMPERES.

NO. OF BEADS - TWO, WITH ALL STARTS AND STOPS STAGGERED. CLEAN SLAG WITH BLUNT TOOL AND STAINLESS STEEL WIRE BRUSH BEFORE CONTINUING A WELD.

ARC TRAVEL SPEED - 1ST LAYER AT 10" PER MINUTE.
2ND LAYER AT 9" PER MINUTE.

HEAT TREATMENT - NONE

NOTES:

THE ELECTRODE AND PROCEDURE RECOMMENDED WILL YIELD A CORROSION RESISTANT, LOW FLAT WELD WITH MINIMUM PENETRATION, FOR EASE OF LATER REMOVAL BY GRINDING. TWO LAYERS ARE USED TO INSURE AGAINST POROSITY LEAKAGE. THE "AHEAD OF CENTER" POSITION SHOULD NOT BE EXCEEDED AS MOLTEN METAL AND SLAG MAY RUN AHEAD OF ARC AND CAUSE A FAULTY WELD.

FIGURE 19

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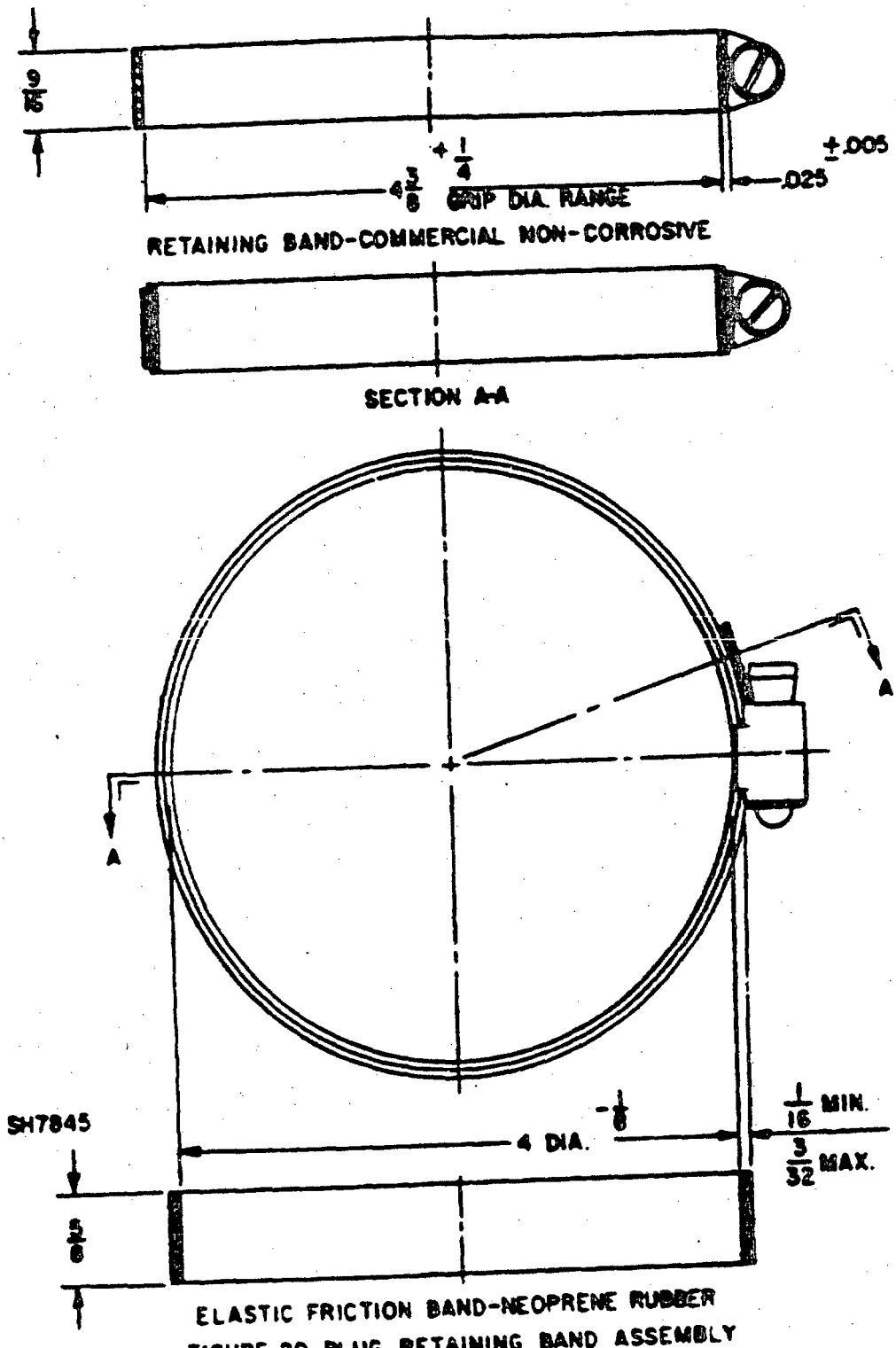


FIGURE 20 PLUG RETAINING BAND ASSEMBLY

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APPENDIX

10. REQUIREMENTS FOR WELDING QUALIFICATION

10.1 SCOPE

10.1.1 This appendix covers the minimum requirements necessary to qualify welding procedures; welding equipment; welders and welding operators.

20. General requirements.-

20.1 The purpose of these qualification requirements is to insure that proven procedures are used with adequate equipment by properly trained personnel.

20.2 It shall be the responsibility of the supplier to perform the necessary qualification tests prior to the start of production work.

20.3 It shall be the responsibility of the supplier to maintain the record of the tests required and if requested provide a record of test results to the Bureau of Ships or its authorized representative.

20.4 It shall be the responsibility of the supplier to insure that only qualified personnel, procedures, and equipment are used.

21. Welding procedure qualification.-

21.1 Destructive and non-destructive tests shall be used for qualification of welding procedures.

21.1.1 Destructive tests: Five types of destructive tests shall be used as required by this section:

(a) Reduced section tension tests
(b) Side bend tests
(c) Charpy V-notch impact tests
(d) Macro-etch specimens
(e) Explosion bulge tests

21.1.1.2 Reduced section tension tests shall be prepared in accordance with table 4 and figures 21 and 22 and shall be performed in accordance with MIL-STD-418.

21.1.1.3 Side bend test specimens shall be prepared in accordance with table 4 and figures 22 and 23 and shall be performed in accordance with MIL-STD-418.

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21.1.1.4 Charpy V-notch impact tests shall be performed in accordance with the applicable electrode or material specifications and prepared in accordance with figure 22.

21.1.1.5 Macro-etch specimens shall be suitably prepared and etched to reveal cross sectional soundness.

21.1.1.6 Explosion bulge tests shall be performed as required by the Bureau of Ships.

21.1.1.7 Non-destructive tests: Non-destructive tests shall consist of the following, where applicable, and shall be prepared and performed in accordance with MIL-STD-271 and inspected to standards as outlined in 21.2.2.3.

(a) Radiographic examination
(b) Magnetic particle examination
(c) Dye penetrant examination

21.2 Procedure qualification.- Procedure qualification tests shall be made using specified base material with the production standard weld joint. The test plate thickness shall be the maximum used in production.

21.2.1 Qualification level number 1.-

21.2.1.1 Qualification level number 1 covers any procedure to be qualified using materials, processes, or techniques that are detailed in this document.

21.2.1.2 Qualification test plates shall be of sufficient size to prepare and perform all of the destructive and non-destructive tests outlined in 21.1.1 and 21.1.1.7 and prepare according to table IV and figures 21 through 24.

Table IV - Type and number of test specimens and range of thickness qualified.

Thickness, (t), of test plate	Range of thickness of material qualified by test plate		Type and number of test required	
	Minimum	Maximum	Reduced section	Side bend
3/4 inches and over	3/16 inch	2t	2	3

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21.2.1.3 Qualification level number 1 procedure: Record on a form in accordance with figure 25 at least the following information:

(a) Base material
(b) Filler metal
(c) Base material preparation
(d) Joint design
(e) Welding process
(f) Electrical characteristics
(g) Welding position
(h) Shielding gas
(i) Heat treatments
(j) Non-destructive test results
(k) Destructive test results

21.2.2 Qualification level number 2.-

21.2.2.1 Qualification level number 2 is intended to permit procedure approval for a variation in procedure which has passed a qualification level number 1 procedure test.

21.2.2.2 A separate qualification test shall be performed for each of the following variables:

(a) Twin arc welding.
(b) A change in welding position.
(c) Any special joints or any joint design with included angle or root opening less than standard joint designs.
(d) A change to a larger electrode size.

21.2.2.3 Test plates shall be at least 15 inches long and shall be subjected to non-destructive or destructive tests as listed below, to prove that the activity is capable of applying the requirements of this document to produce sound welds.

(a) Radiographic examination of at least 12 inches of the length of test weld to acceptance standards NAVSHIPS 250-692-2.
(b) Magnetic particle examination to prove free of lineal disposed discontinuity.
(c) Two macro-etch specimens in accordance with 21.1.1.5 of this appendix where radiography is not applicable.

21.2.2.3.1 Special joint test plates shall be according to production design.

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22. Welder and welding operator qualification.-

22.1 Each welder or welding operator who is to perform production welding on the specified material shall be qualified prior to performing any production welding.

22.2 Each welder, using specified electrodes, or welding wire in accordance with the procedure specified, bead techniques and heat input, shall make a test plate in accordance with figures 26 and 27 and positions in accordance with table V and figure 28 in material the maximum thickness or over to be used in production. Material for test plates shall be the same as used in production. Test plates shall be prepared to conform to figures 26 and 27 of this appendix. A welder qualification record shall certify compliance with bead techniques and heat inputs in accordance with those specified in the procedure.

Table V - Qualification levels number 1 and 2 and welder qualification test and conditions.

Material size	Position in which welding shall be done		
	Flat, vertical, horizontal and O.H. grooves and fillets	Flat and vertical grooves and fillets horizontal fillets	Flat grooves and fillets, horizontal fillets
3/4 inch or over	Figure 22 1/ Figure 26 2/ (In vertical and O.H. positions see figure 28)	Figure 22 1/ Figure 26 2/ (In vertical position see figure 28)	Figure 22 1/ Figure 26 2/ Figure 27 3/ (In flat position see figure 28)

1/ Test plate figure 22 for qualification levels number 1 and 2. Procedure test.

2/ Test plate figure 26 for metallic arc performance test.

3/ Test plate figure 27 for metallic inert gas performance test.

22.3 Welder and welding operator qualification tests shall be examined radiographically and by side bend testing.

22.3.1 Radiographic acceptance shall be based on the standards as outlined in NAVSHIPS 250-692-2.

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22.3.2 Any side bend specimen, in which a crack or opening exists before bending or results from the bending in excess of 1/8 inch in any direction, shall be rejected.

22.4 Welders who perform satisfactory welding procedure qualification tests shall be considered qualified to perform production welding without further re-qualification.

22.5 Retests.-

22.5.1 If any of the welders qualification test welds fails to meet the above requirements, he may be allowed a retest under one of the following conditions:

(a) An immediate retest may be made, that shall consist of two test welds of each type he failed, all of which shall meet the test requirements above.
(b) A retest may be made provided the welder has had further training or practice, in which case a complete retest should be made for each position in which the welder failed to meet the requirements.

22.6 Each qualified welder or welding operator who is to use any special techniques or variation of techniques within the basic procedure, such as twin arc welding, shall make up a test plate at least 15 inches long duplicating the special conditions. Results shall be checked by radiography or by Macro-Etch specimen where radiography is not applicable to prove their ability to deposit sound welds.

22.7 Requalification of welders or welding operators.-

22.7.1 A welder or welding operator must be requalified whenever one or more of the changes listed below are made in performance procedure.

(a) Addition of approved welding positions other than those he is already qualified for.
(b) A change from one welding process to another for which he is not qualified.
(c) A welder shall be requalified if his work is considered unsatisfactory.
(d) A welder shall be requalified if he does not use the process for which he was qualified for a period of 3 months or more.

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23. Welding equipment qualification.-

23.1 Welding equipment qualification tests in general are not required for equipment that has been used in production and proven capable of producing proper welding characteristics. If, for any reason, any equipment is suspect, it shall be proven to the inspector that it is capable of producing acceptable welds when employed by a qualified operator and using a qualified procedure. Equipment not capable of producing acceptable welds shall not be used.

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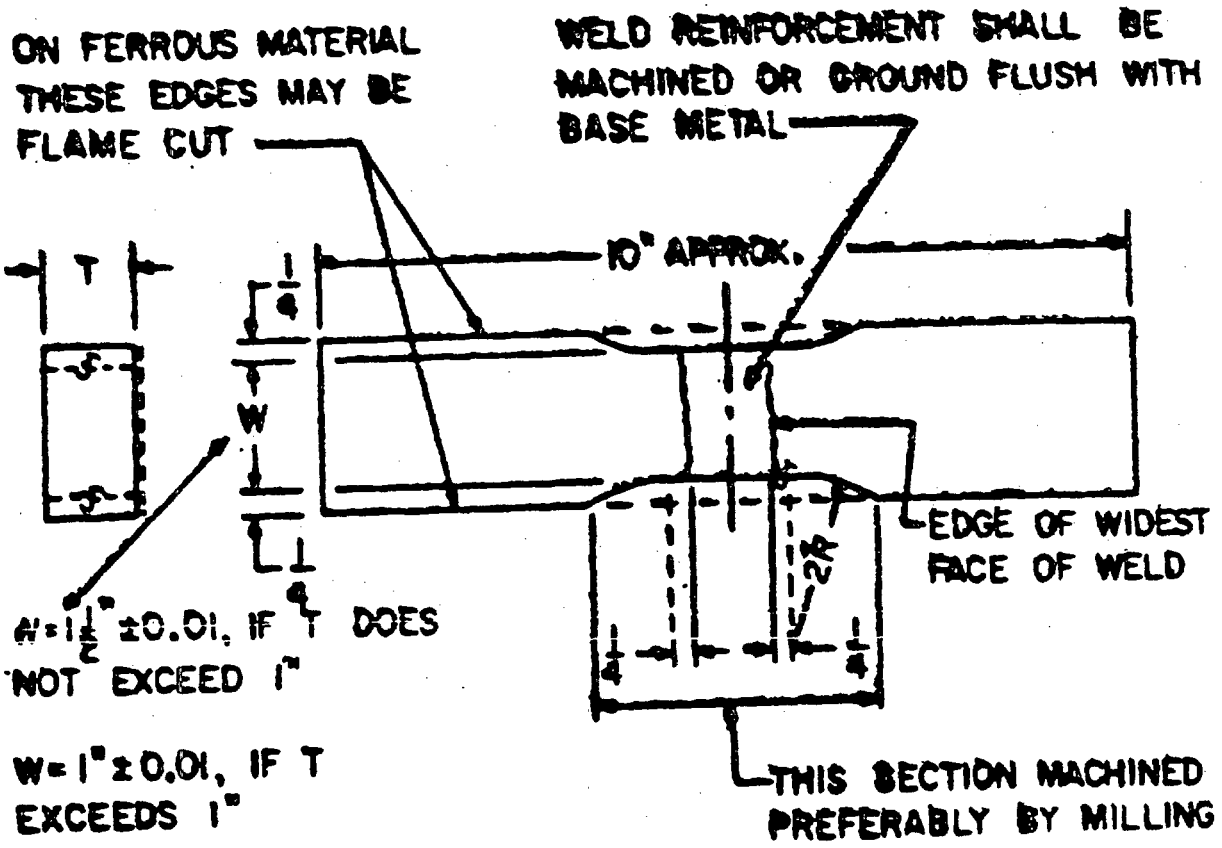
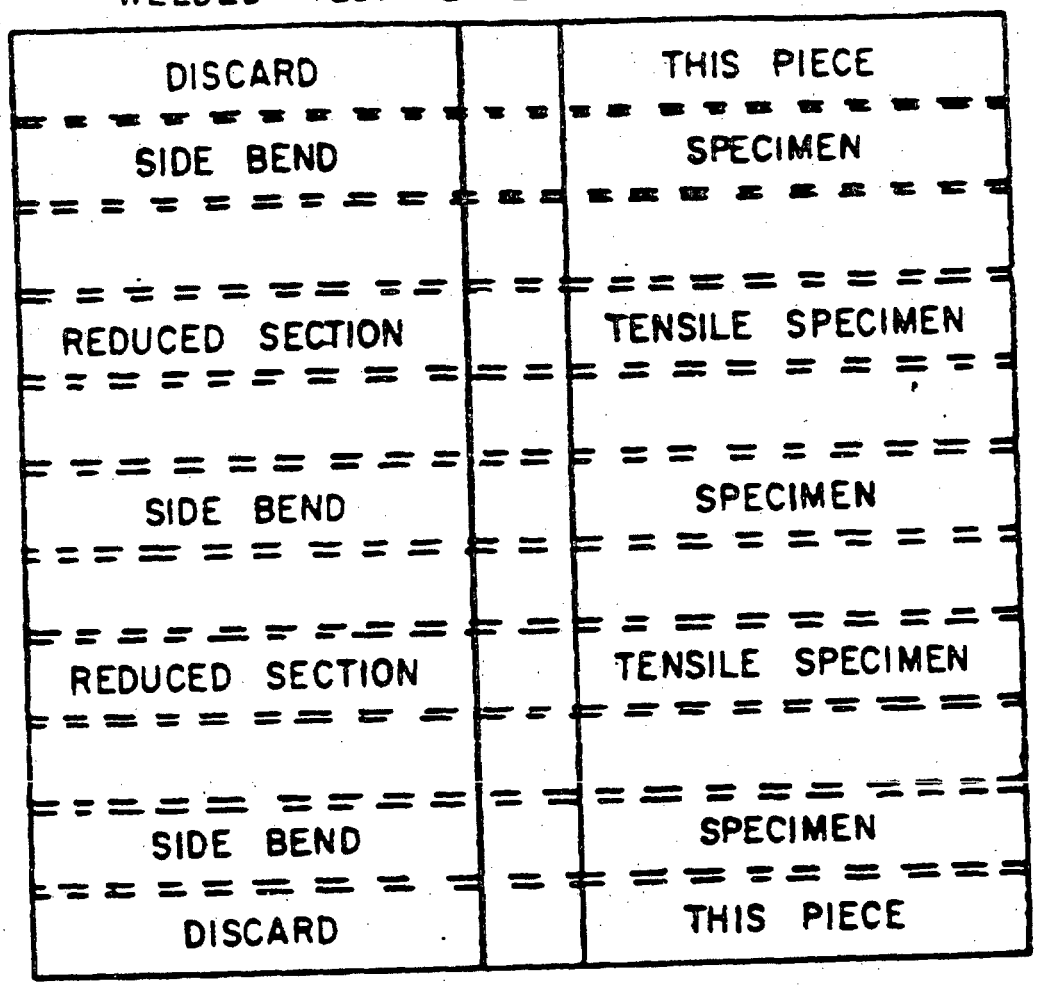


FIGURE 21

REDUCED-SECTION TENSION SPECIMEN PLATE

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ORDER OF REMOVAL OF ~~TEST~~ SPECIMENS FROM WELDED TEST PLATE OVER 3/4" THICK.



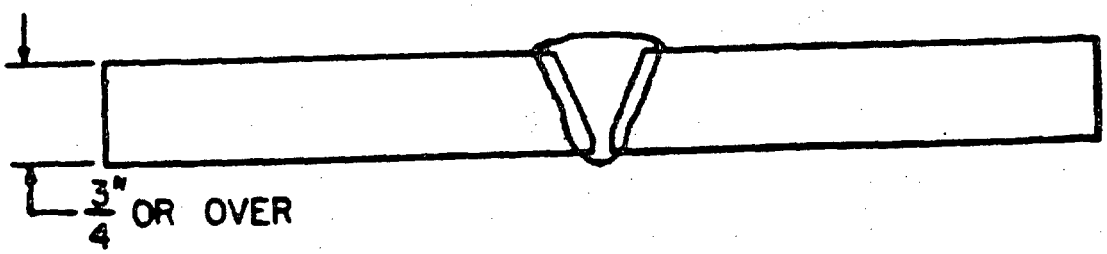
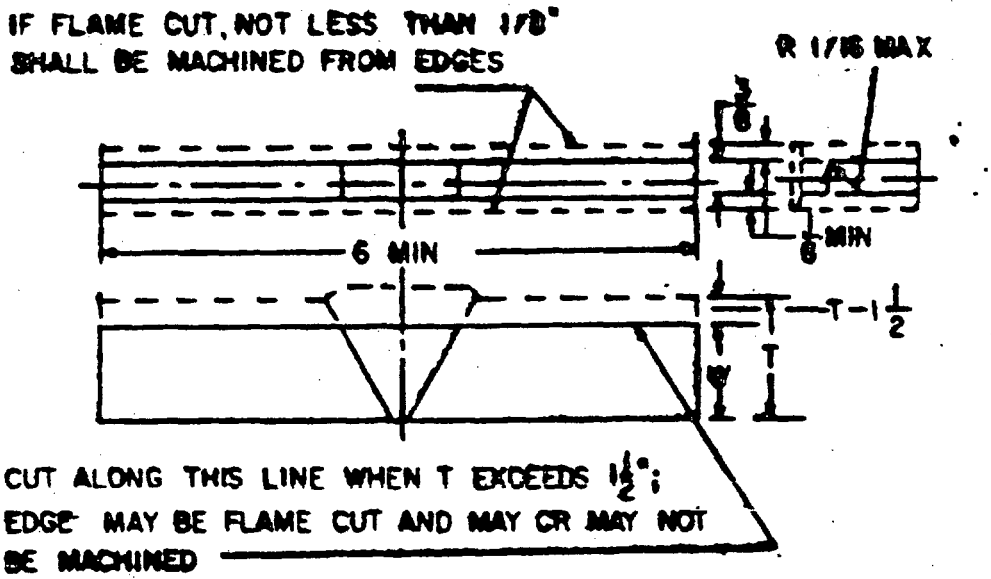


FIGURE 22

SR7204

LOCATION OF TENSION AND BEND SPECIMENS

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T, IN.	W, IN.
3/4 TO 1-1/2	T
>1-1/2	NOTE 1

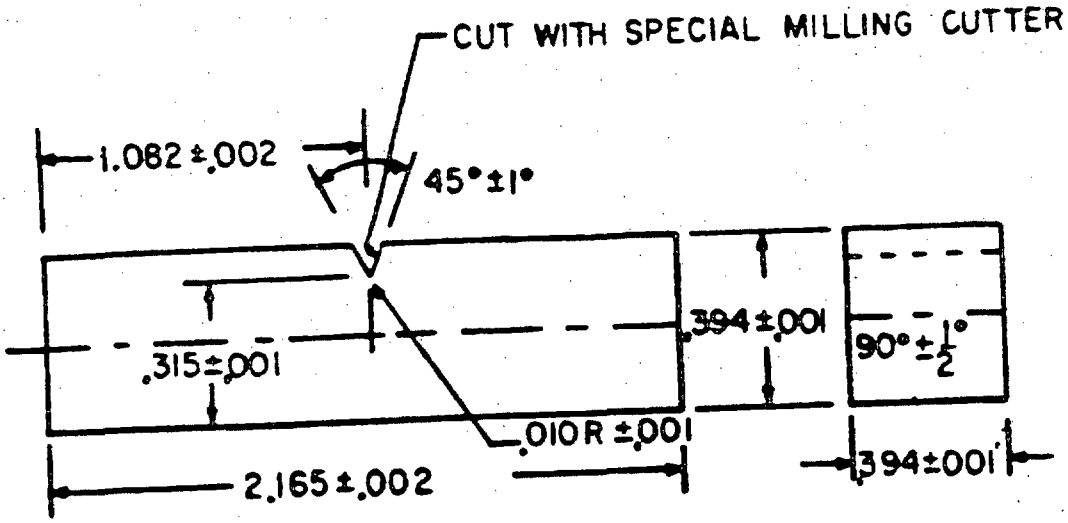
NOTE:

1. FOR PLATE OVER 1-1/2 INCHES THICK, CUT SPECIMEN INTO APPROXIMATELY EQUAL STRIPS BETWEEN 3/4-INCH AND 1/2 INCH WIDE, AND TEST EACH STRIP.

FIGURE 23

SIDE-BEND SPECIMEN

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NOTES:

1. CUT NOTCH WITH SPECIAL MILLING CUTTER.
2. MAKE FOUR CUTS PER NOTCH IN FOLLOWING ORDER: (A) .040 (B) .025 (C) .010 (D) .004
3. NO IRREGULARITIES IN CONTOUR OF APEX OF NOTCH.
4. NO. 8 POLISH MICRO-INCH FINISH.
5. WHEN USED FOR DEPOSITED WELD METAL, THE V-NOTCH SHALL BE LOCATED NORMAL TO THE SURFACE OF THE WELD.

FIGURE 24
V-NOTCHED CHARPY SPECIMEN.

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Specification No..... Date.....
Welding Process..... Manual or machine.....
Material Specification.....to.....
Thickness .....
Thickness Range this test qualifies.....
Filler Metal..... Flux or Atmosphere
Flux trade name or composition.....
Inert Gas Composition.....
Trade Name.....Flow Rate.....
Is Backing Strip used?.....
Preheat Temperature Range.....

Welding Procedure

Single or Multiple Pass.....Postheat Treatment.....
Single or Multiple Arc.....
Position of groove.....
(Flat, horisontal, vertical, or overhead, if vertical, state whether upward or downward)

For Information Only

Filler Wire Diameter.....
Trade Name..... Welding Techniques
Type of Backing..... Joint dimensions accord with.....
Forehand or Backhand..... amps....volts....inches per min...

Reduced Section Tensile Test Figure 21

Spec. No.	Dimensions		Area	Ultimate Total Load, Lb.	Ultimate Unit Stress p.s.i.	Character of Failure and location
	Width	Thickness				

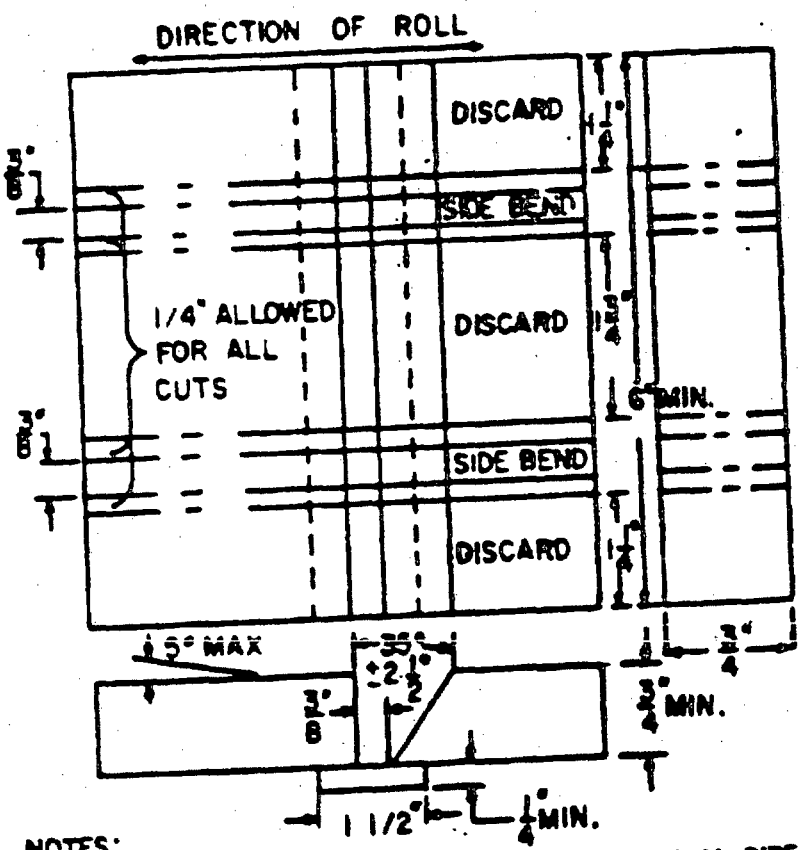
Charpy Impact and Guided Bend Tests Figures 23 and 24

Type & Figure No.	Result	Type & Figure No.	Results

Welders name..... Clock No.....
This by virture of these tests meets welder performance requirements.
Test Conducted by..... Laboratory Test No.....
per.....
Signed.....
(Manufacturer)
Date..... By.....

Figure 25 - Manufacturer's record of welding procedure qualification tests.

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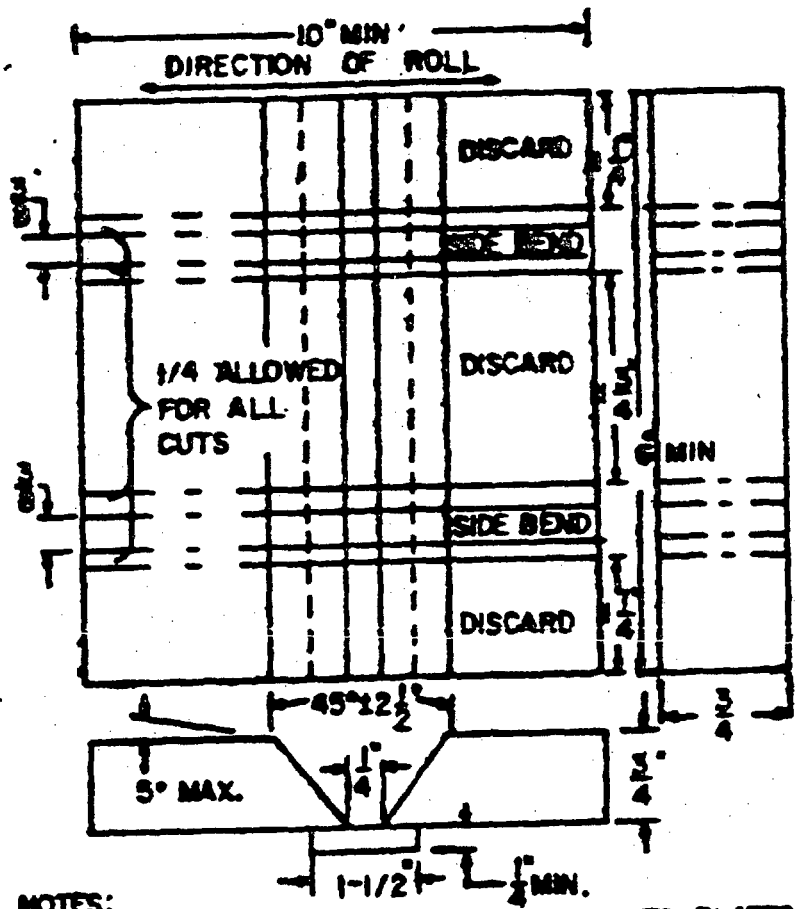
NOTES:

1. WELDS SHALL BE MADE USING THE MAXIMUM SIZE OF ELECTRODE OR WELDING ROD SUITABLE FOR THE POSITION AND THICKNESS OF TEST PLATE.
2. BACKING STRAP SHALL BE CONTIGUOUS TO PLATES.
3. JOINTS WELDED IN THE VERTICAL POSITION SHALL BE WELDED UPWARDS.
4. WELDING SHALL BE DONE FROM ONE SIDE ONLY.
5. MACHINE REINFORCEMENT AND BACKING STRAP FLUSH. DO NOT REMOVE ANY UNDERCUTTING.
6. SPECIMENS MAY BE MACHINED OR OXYGEN-CUT FROM PLATE.
7. BREAK EDGES OF SPECIMENS TO A RADIUS OF T/6 MAXIMUM.

FIGURE 26

WELDERS QUALIFICATION TEST FOR MANUAL METALLIC ARC

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NOTES:
1. BACKING STRAP SHALL BE CONTIGUOUS TO PLATES.
2. WELDING SHALL BE DONE FROM ONE SIDE ONLY.
3. MACHINE REINFORCEMENT AND BACKING STRAP FLUSH. DO NOT REMOVE ANY UNDERCUTTING.
4. SPECIMENS MAY BE MACHINED OR OXYGEN-CUT FROM PLATE.
5. BREAK EDGES OF SPECIMENS TO A RADIUS OF T/6 MAX.

FIGURE 27

WELDER QUALIFICATION TEST PLATE
(INERT GAS SHIELDED METALLIC ARC)

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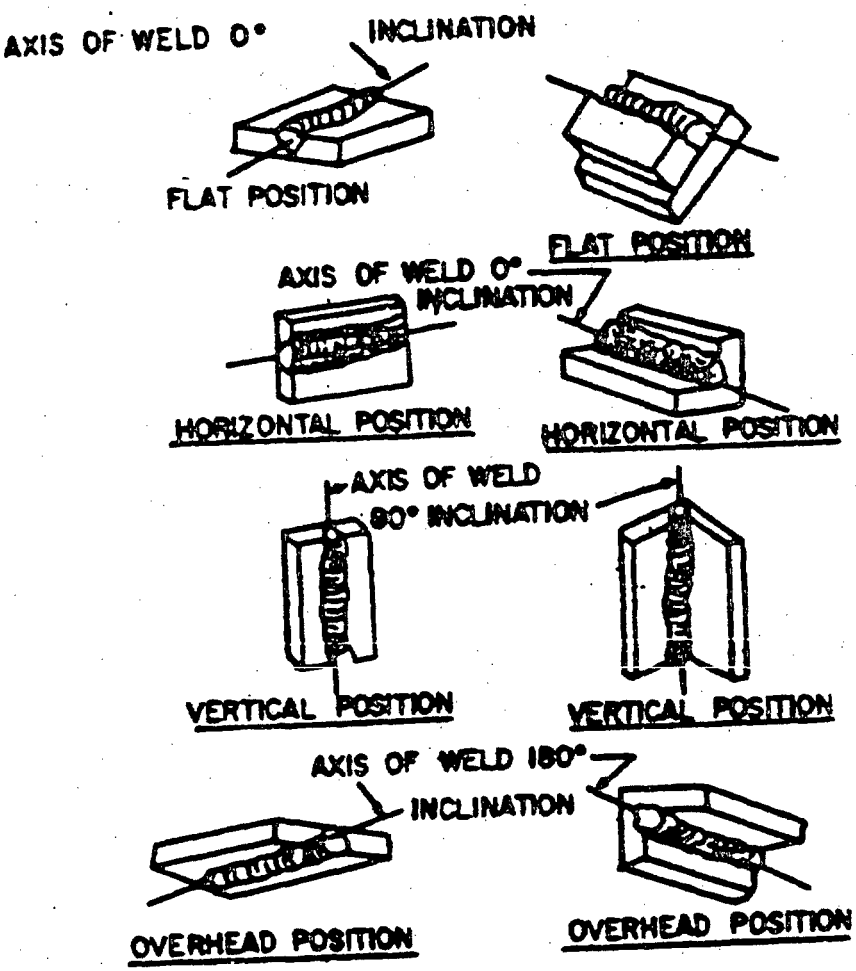


FIGURE 28
POSITION OF WELDING

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that available products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be accomplished. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

   ☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity) | DATE

DD FORM 1 APR 68 1426 REPLACES NAVSHIPS FORM 4853, WHICH IS OBSOLETE

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